Differential Equations Workbook Solutions

Modeling with differential equations

krista king MATH

DIRECTION FIELDS AND SOLUTION CURVES

■ 1. Sketch the direction field of the differential equation.

$$y' = 2y + x$$

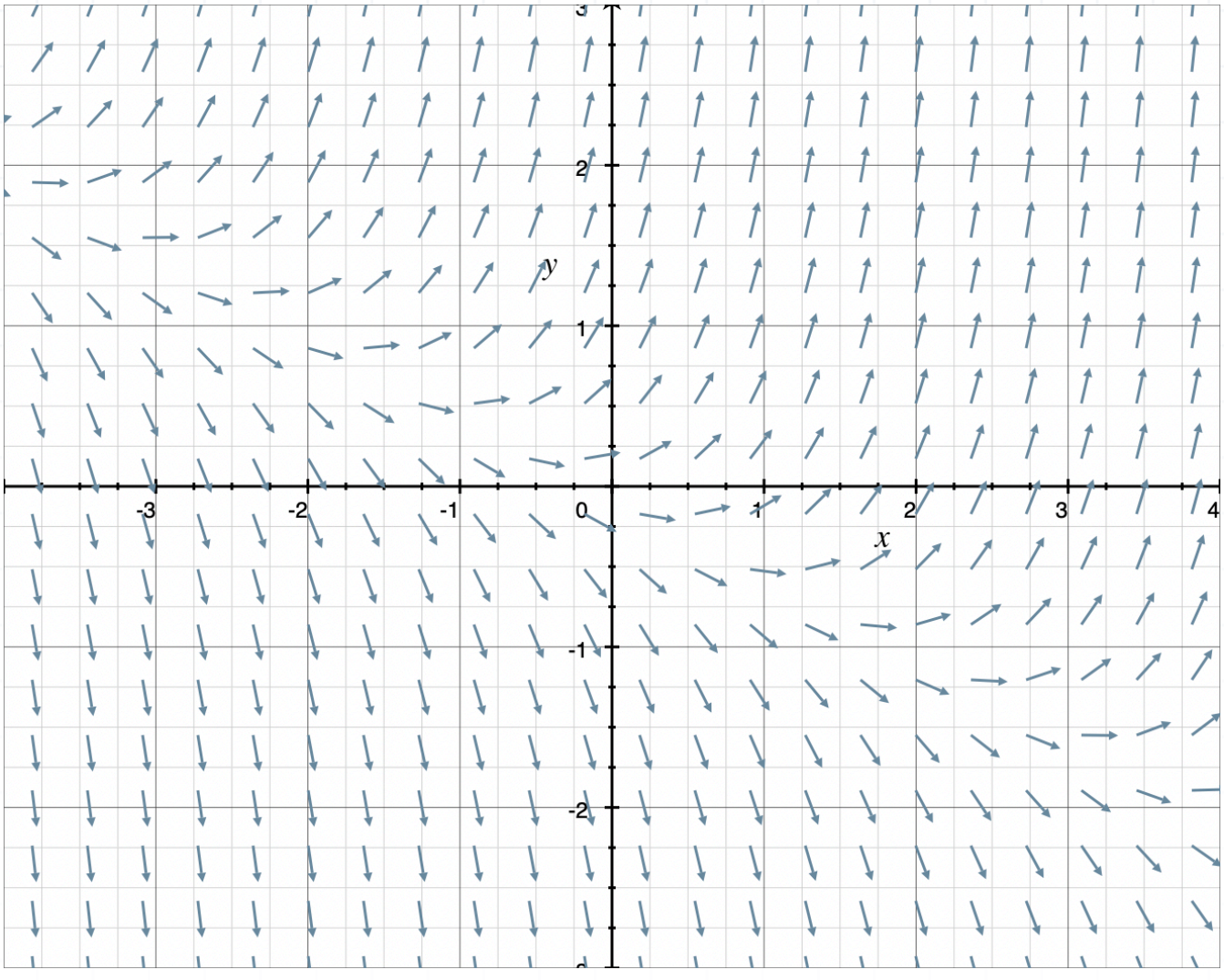
Solution:

We'll make a table of values of y' for every combination of $x = \{-2, -1, 0, 1, 2\}$ and $y = \{-2, -1, 0, 1, 2\}$.

		x				
		-2	**-1**	**0**	**1**	**2**
y	**-2**	-6	-5	-4	-3	-2
	-1	-4	-3	-2	-1	0
	0	-2	-1	0	1	2
	1	0	1	2	3	4
	2	2	3	4	5	6

Plotting each of these values as a tangent, for instance plotting the slope -6 at the point $(-2, -2)$, gives us the beginning of a direction field. And if we keep adding more and more tangents to get finer detail, the direction field should look something like this:

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■ 2. Sketch the direction field of the differential equation, and the solution curve passing through $(1,1)$.

$$y' = -3y + x$$

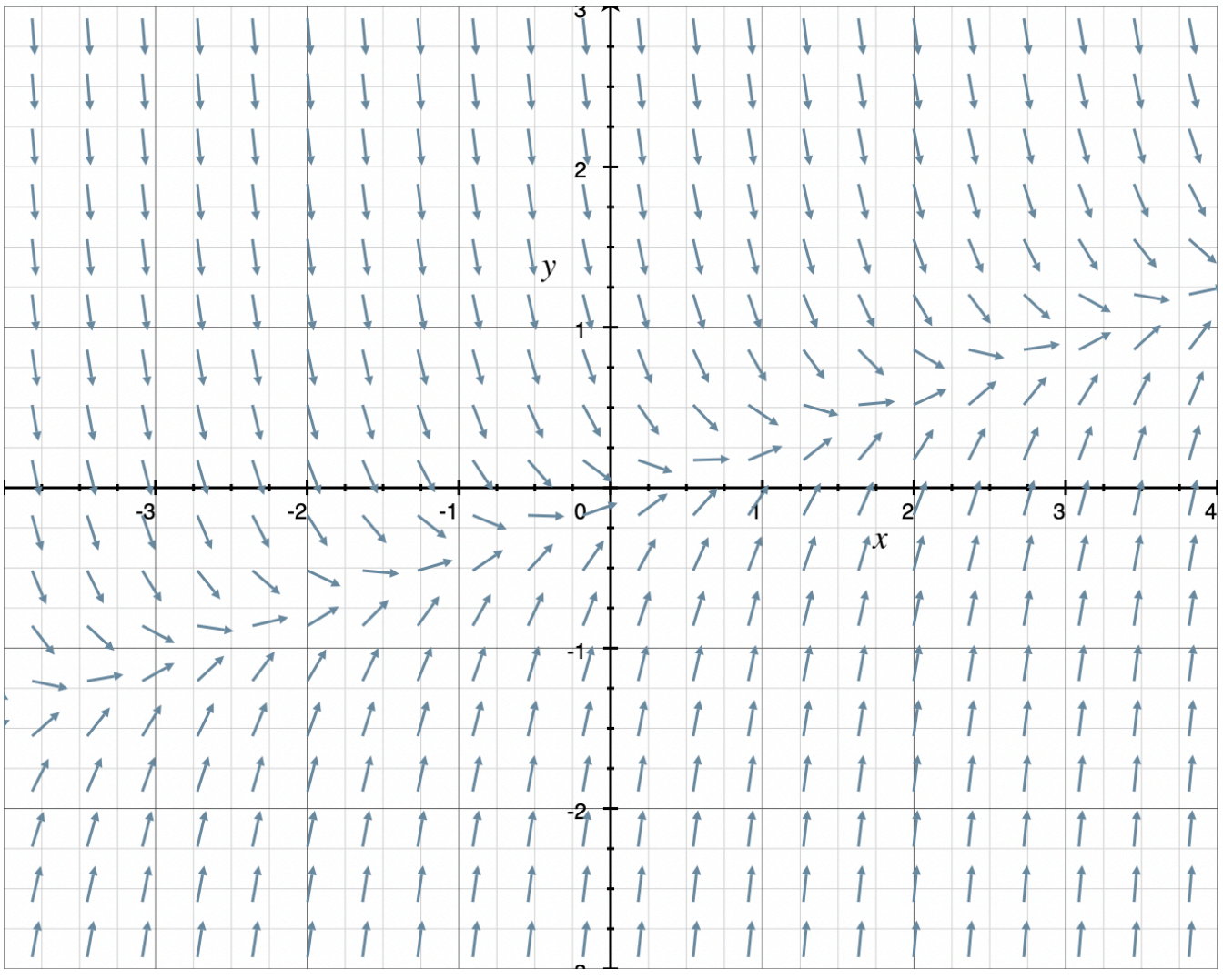
Solution:

We'll make a table of values of y' for every combination of $x = \{-2, -1, 0, 1, 2\}$ and $y = \{-2, -1, 0, 1, 2\}$.

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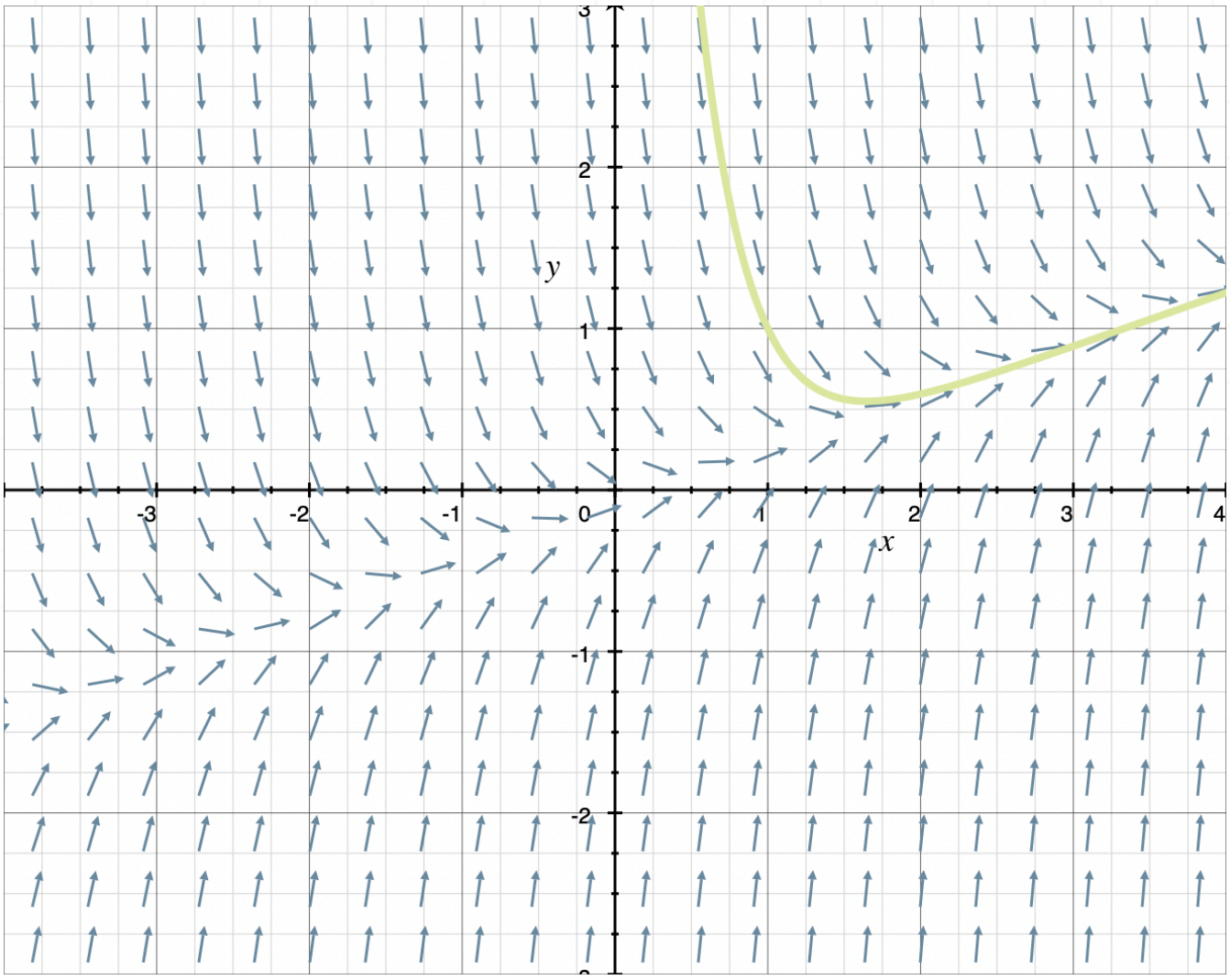
		x				
		-2	**-1**	**0**	**1**	**2**
y	**-2**	4	5	6	7	8
	-1	1	2	3	4	5
	0	-2	-1	0	1	2
	1	-5	-4	-3	-2	-1
	2	-8	-7	-6	-5	-4

Plotting each of these values as a tangent, for instance plotting the slope 4 at the point $(-2, -2)$, gives us the beginning of a direction field. And if we keep adding more and more tangents to get finer detail, the direction field should look something like this:



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Following the tangents in the direction field around $(1,1)$, we can approximate the solution curve through that point.



■ 3. Sketch the direction field of the differential equation, and the solution curve passing through $(0,0)$.

$$y' = y^2 + x^2 - 1$$

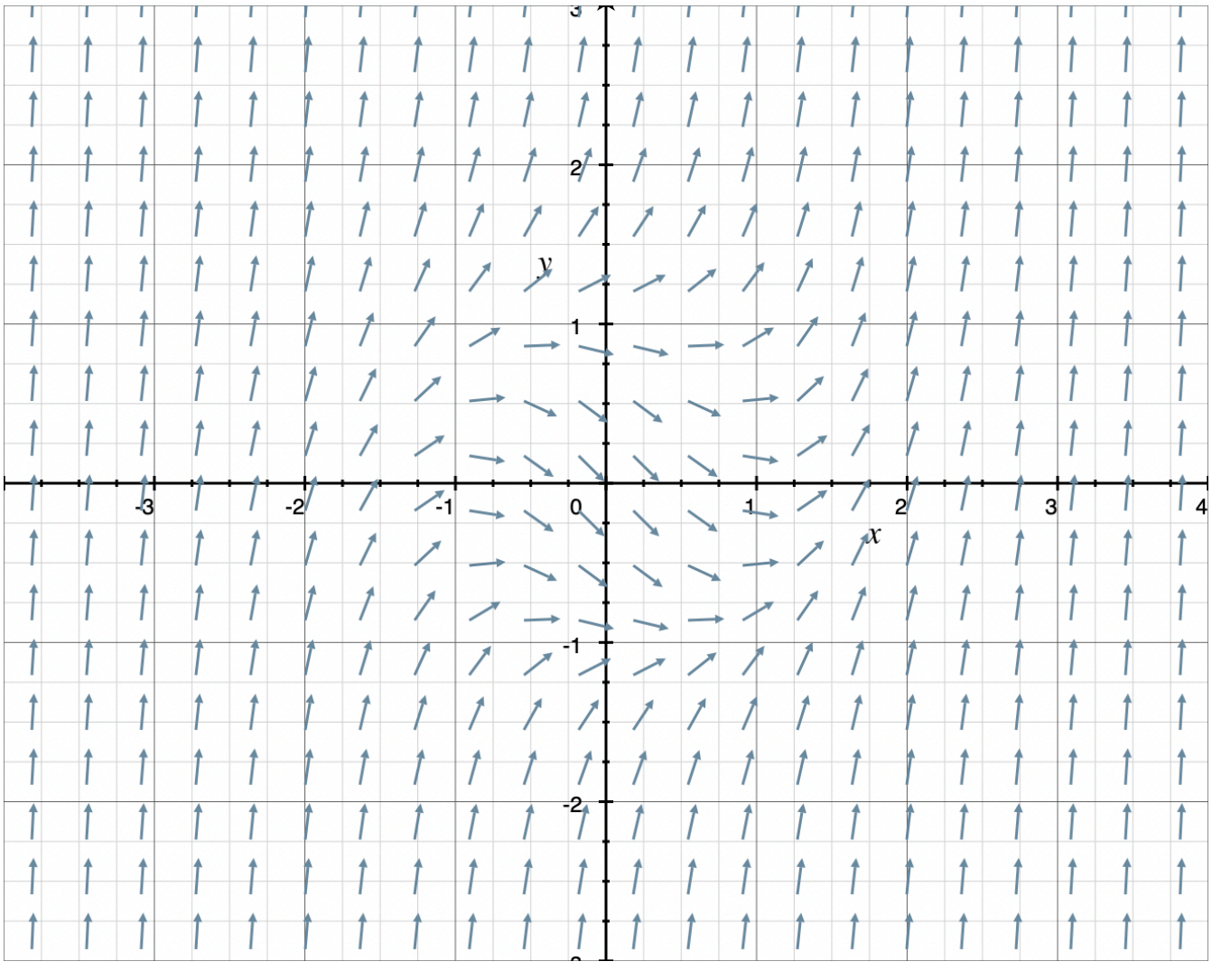
Solution:

We'll make a table of values of y' for every combination of $x = \{-2, -1, 0, 1, 2\}$ and $y = \{-2, -1, 0, 1, 2\}$.

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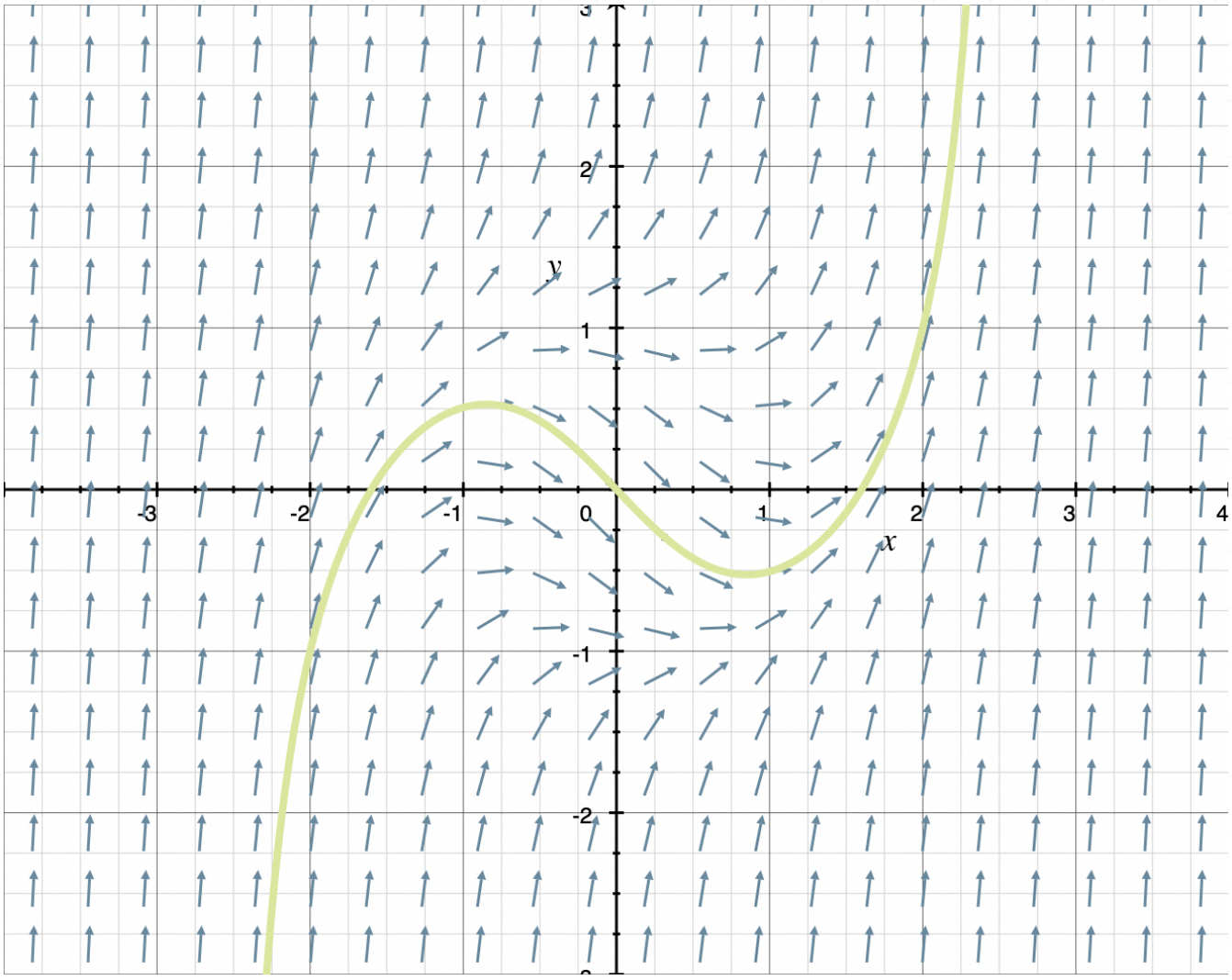
		x				
		-2	-1	0	1	2
y	-2	7	4	3	4	7
	-1	4	1	0	1	4
	0	3	0	-1	0	3
	1	4	1	0	1	4
	2	7	4	3	4	7

Plotting each of these values as a tangent, for instance plotting the slope 7 at the point $(-2, -2)$, gives us the beginning of a direction field. And if we keep adding more and more tangents to get finer detail, the direction field should look something like this:



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Following the tangents in the direction field around $(0,0)$, we can approximate the solution curve through that point.



■ 4. Sketch the direction field for the differential equation, and the solution curve passing through $(0,1)$.

$$y' = 2y^2 - x^2 - 2$$

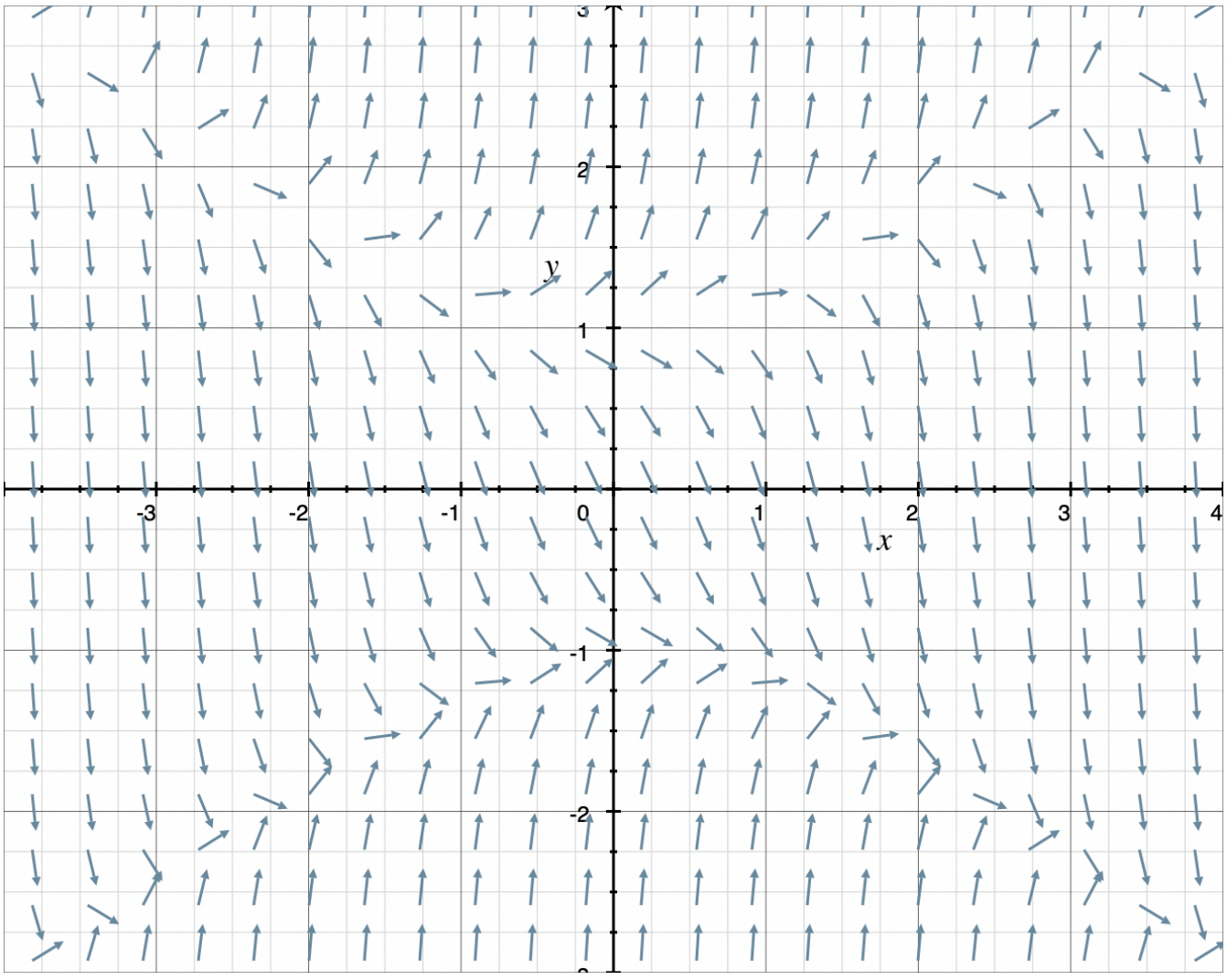
Solution:

We'll make a table of values of y' for every combination of $x = \{-2, -1, 0, 1, 2\}$ and $y = \{-2, -1, 0, 1, 2\}$.

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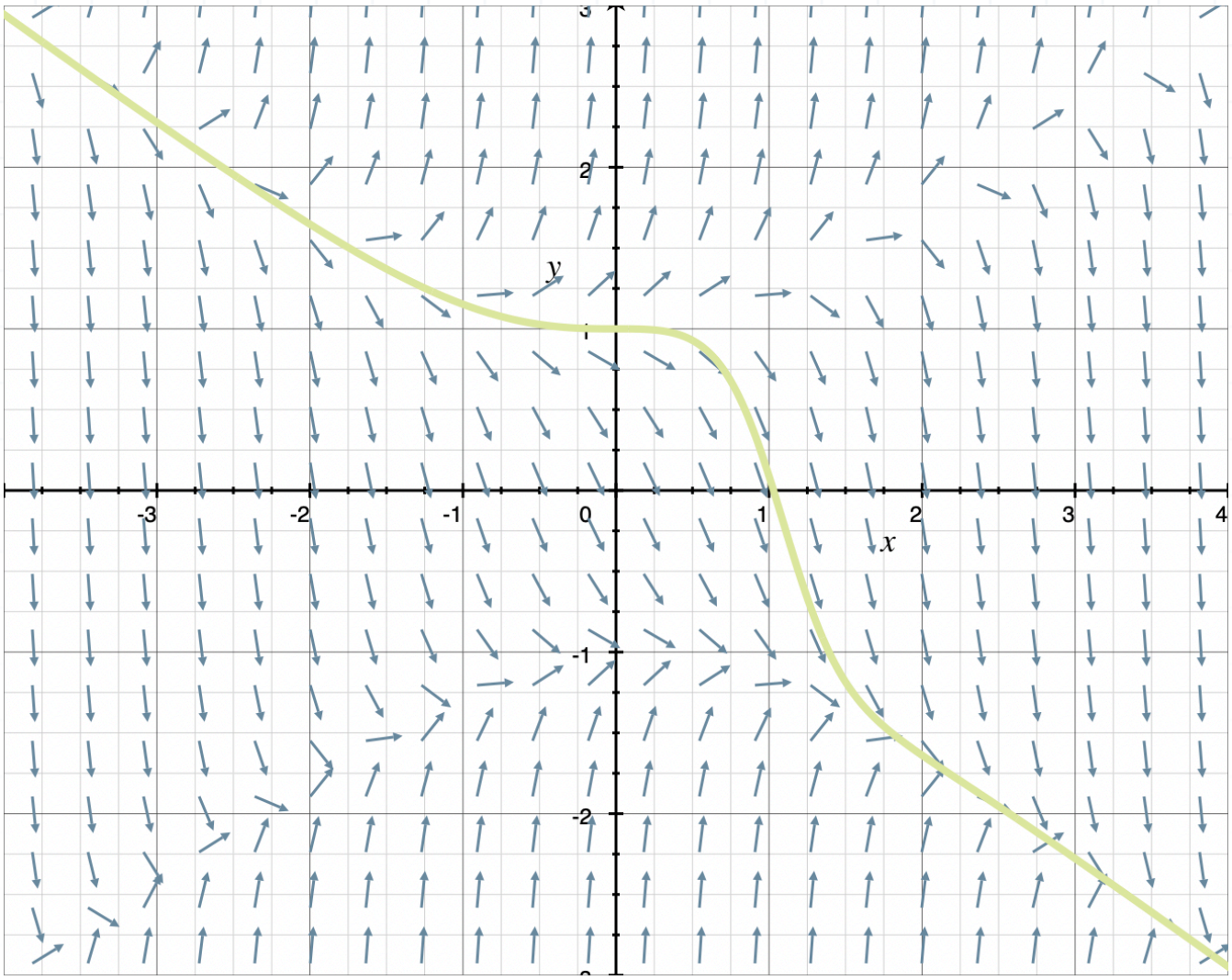
		x				
		-2	**-1**	**0**	**1**	**2**
y	**-2**	2	5	6	5	2
	-1	-4	-1	0	-1	-4
	0	-6	-3	-2	-3	-6
	1	-4	-1	0	-1	-4
	2	2	5	6	5	2

Plotting each of these values as a tangent, for instance plotting the slope 2 at the point $(-2, -2)$, gives us the beginning of a direction field. And if we keep adding more and more tangents to get finer detail, the direction field should look something like this:



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Following the tangents in the direction field around $(0,1)$, we can approximate the solution curve through that point.



5. Sketch the direction field of the differential equation.

$$y' = \sin y + x$$

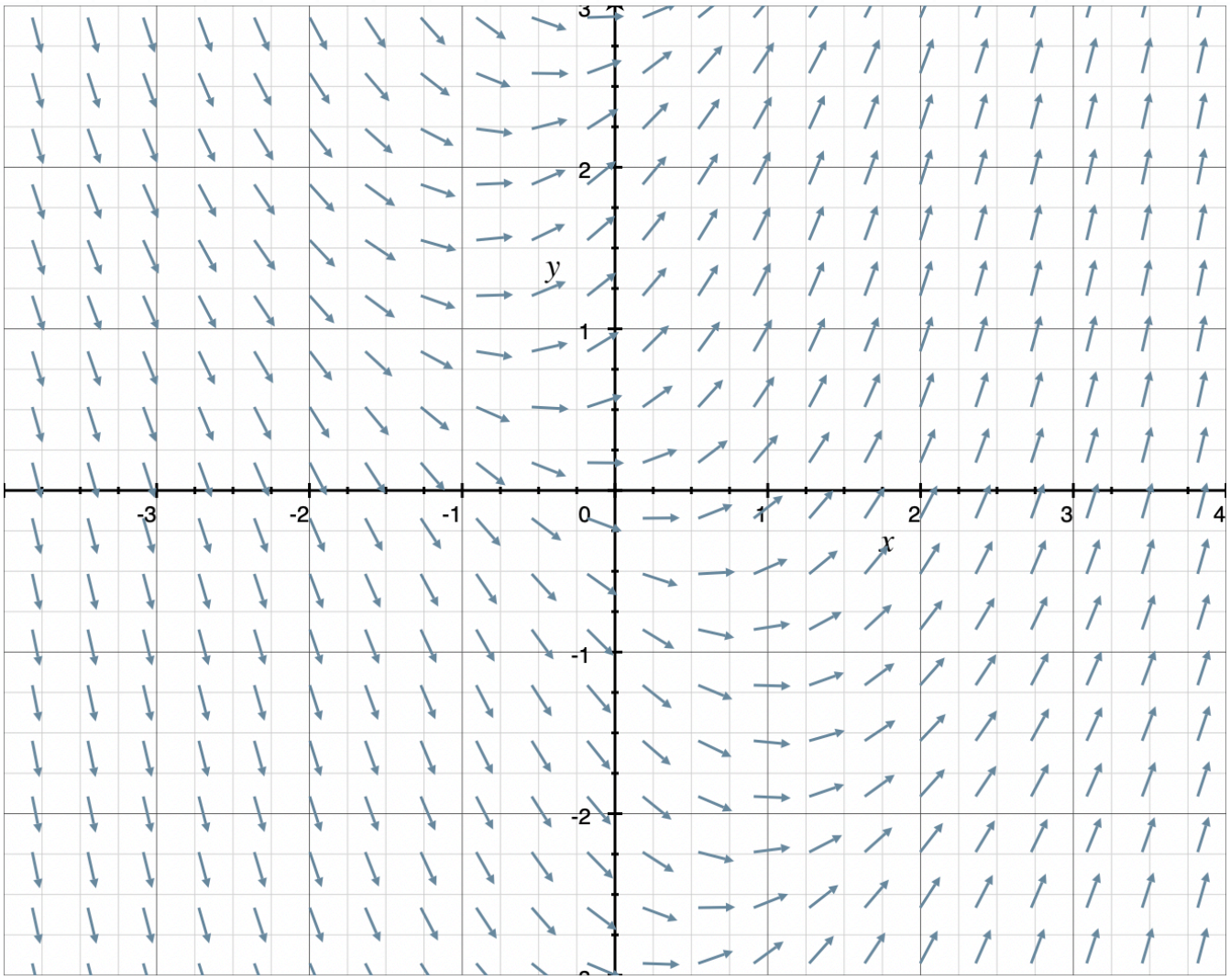
Solution:

We'll make a table of values of y' for every combination of $x = \{-2, -1, 0, 1, 2\}$ and $y = \{-\pi, -\pi/2, 0, \pi/2, \pi\}$.

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		x				
		-2	**-1**	**0**	**1**	**2**
y	**-2**	-2	-1	0	1	2
	-1	-3	-2	-1	0	1
	0	-2	-1	0	1	2
	1	-1	0	1	2	3
	2	-2	-1	0	1	2

Plotting each of these values as a tangent, for instance plotting the slope -2 at the point $(-2, -2)$, gives us the beginning of a direction field. And if we keep adding more and more tangents to get finer detail, the direction field should look something like this:



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6. Sketch the direction field of the differential equation.

$$y' = xy$$

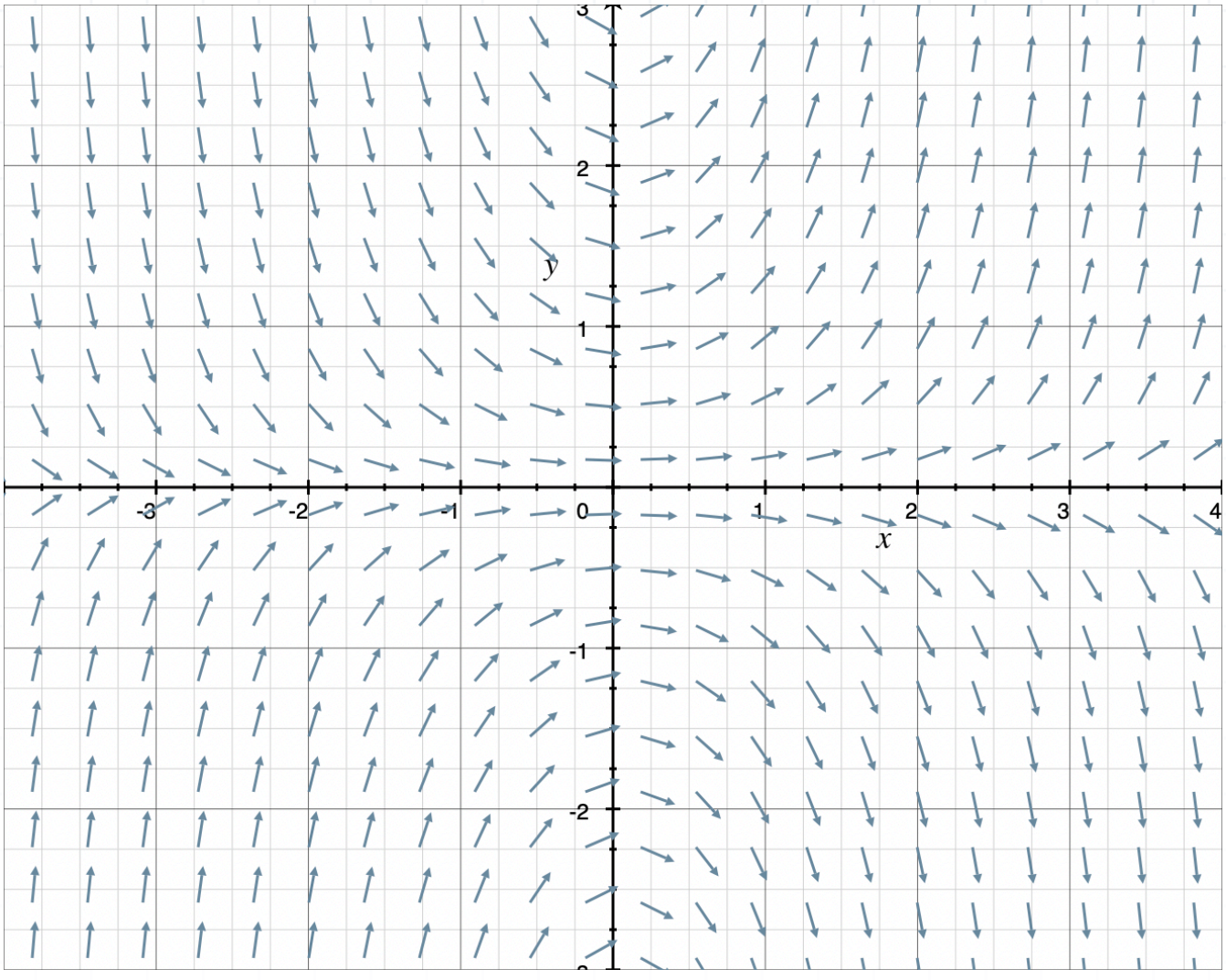
Solution:

We'll make a table of values of y' for every combination of $x = \{-2, -1, 0, 1, 2\}$ and $y = \{-2, -1, 0, 1, 2\}$.

		x				
		-2	**-1**	**0**	**1**	**2**
y	**-2**	4	2	0	-2	-4
	-1	2	1	0	-1	-2
	0	0	0	0	0	0
	1	-2	-1	0	1	2
	2	-4	-2	0	2	4

Plotting each of these values as a tangent, for instance plotting the slope 4 at the point $(-2, -2)$, gives us the beginning of a direction field. And if we keep adding more and more tangents to get finer detail, the direction field should look something like this:

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INTERVALS OF VALIDITY

■ 1. Find the interval of validity for the solution to the differential equation, given $y(3) = -4$.

$$(x^2 - 25)y' - 7y - \ln(10 - x) = 0$$

Solution:

Put the equation into standard form for a linear equation.

$$(x^2 - 25)y' - 7y - \ln(10 - x) = 0$$

$$\frac{dy}{dx} - \frac{7y}{x^2 - 25} - \frac{\ln(10 - x)}{x^2 - 25} = 0$$

$$\frac{dy}{dx} - \frac{7}{x^2 - 25}y = \frac{\ln(10 - x)}{x^2 - 25}$$

From this new form, we can identify

$$P(x) = -\frac{7}{x^2 - 25}$$

$$Q(x) = \frac{\ln(10 - x)}{x^2 - 25}$$

both of which are undefined at

$$x^2 - 25 = 0$$

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$$x^2 = 25$$

$$x = \pm 5$$

The functions $P(x)$ and $Q(x)$ are undefined for

$$10 - x < 0$$

$$x > 10$$

So we need to keep $x < 10$, which means we'll only consider the interval $(-\infty, 10)$. The values $x = \pm 5$ divide this interval into three possible intervals of validity.

$$-\infty < x < -5 \qquad -5 < x < 5 \qquad 5 < x < 10$$

Because the initial condition is $y(3) = -4$, and the value $x_0 = 3$ is contained within $(-5,5)$, we can say that $(-5,5)$ is the interval of validity.

■ 2. Find the interval of validity for the solution to the differential equation, given $y(0) = 1$.

$$(x+2)y' + xye^{3x} - 5 = 0$$

Solution:

Put the equation into standard form for a linear equation.

$$(x+2)y' + xye^{3x} - 5 = 0$$

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$$\frac{dy}{dx}+\frac{xye^{3x}}{x+2}-\frac{5}{x+2}=0$$

$$\frac{dy}{dx}+\frac{xe^{3x}}{x+2}y=\frac{5}{x+2}$$

From this new form, we can identify

$$P(x)=\frac{xe^{3x}}{x+2}$$

$$Q(x)=\frac{5}{x+2}$$

both of which are undefined at

$$x+2=0$$

$$x=-2$$

This value separates $(-\infty,\infty)$ into two possible intervals of validity.

$$-\infty<x<-2 \qquad\qquad -2<x<\infty$$

Because the initial condition is $y(0)=1$, and the value $x_0=0$ is contained within $(-2,\infty)$, we can say that $(-2,\infty)$ is the interval of validity.

■ 3. Find the interval of validity for the solution to the differential equation, given $y(0)=-1$, and solve the initial value problem to verify that the interval of validity is correct.

$$(x^2-1)y'+2xy=x+1$$

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Solution:

Put the equation into standard form for a linear equation.

$$(x^2-1)y'+2xy=x+1$$

$$\frac{dy}{dx}+\frac{2xy}{x^2-1}=\frac{x+1}{x^2-1}$$

$$\frac{dy}{dx}+\frac{2x}{x^2-1}y=\frac{1}{x-1}$$

From this new form, we can identify

$$P(x)=\frac{2x}{x^2-1}$$

$$Q(x)=\frac{1}{x-1}$$

which together are undefined at

$$x^2-1=0$$

$$x^2=1$$

$$x=\pm 1$$

These two values separate $(-\infty,\infty)$ into three possible intervals of validity.

$$-\infty<x<-1 \qquad -1<x<1 \qquad 1<x<\infty$$

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Because the initial condition is $y(0) = -1$, and the value $x_0 = 0$ is contained within $(-1,1)$, we can say that $(-1,1)$ is the interval of validity.

If we also wanted to find the particular solution by solving the initial value problem, we'd find the integrating factor,

$$\rho(x) = e^{\int P(x)\,dx}$$

$$\rho(x) = e^{\int \frac{2x}{x^2-1}\,dx}$$

$$\rho(x) = e^{\ln(x^2-1)}$$

$$\rho(x) = x^2 - 1$$

multiply through the linear equation,

$$\frac{dy}{dx}(x^2-1) + 2xy = \frac{1}{x-1}(x^2-1)$$

$$\frac{d}{dx}((x^2-1)y) = x + 1$$

integrate both sides,

$$\int \frac{d}{dx}(y(x^2-1))\,dx = \int x + 1\,dx$$

$$(x^2-1)y = \frac{x^2}{2} + x + C$$

and then solve for y to find the general solution.

$$y = \frac{x^2 + 2x + 2C}{2(x^2-1)}$$

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Substituting the initial condition $y(0) = -1$ lets us solve for C,

$$-1 = \frac{0^2 + 2(0) + 2C}{2(0^2 - 1)}$$

$$-1 = \frac{2C}{-2}$$

$$2 = 2C$$

$$C = 1$$

which we can plug back into the general solution to find the particular solution.

$$y = \frac{x^2 + 2x + 2}{2(x^2 - 1)}$$

The solution is undefined where the denominator is 0.

$$x^2 - 1 = 0$$

$$x^2 = 1$$

$$x = \pm 1$$

These two values separate $(-\infty, \infty)$ into three possible intervals of validity.

$$-\infty < x < -1 \qquad -1 < x < 1 \qquad 1 < x < \infty$$

Because the initial condition is $y(0) = -1$, and the value $x_0 = 0$ is contained within $(-1,1)$, we can verify that $(-1,1)$ is the interval of validity.

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■ 4. Find the interval of validity for the solution to the differential equation, given $y(0) = y_0$, for some positive y_0.

$$y' = 3x^2 y^{\frac{4}{3}}$$

Solution:

This is a separable differential equation, which means it's non-linear, and therefore we'll need to find the solution to the equation before we can find the interval of validity. We'll first find the general solution by separating variables,

$$\frac{dy}{dx} = 3x^2 y^{\frac{4}{3}}$$

$$dy = 3x^2 y^{\frac{4}{3}}\, dx$$

$$\frac{1}{y^{\frac{4}{3}}}\, dy = 3x^2\, dx$$

integrating both sides, and then solving for y.

$$\int \frac{1}{y^{\frac{4}{3}}}\, dy = \int 3x^2\, dx$$

$$-\frac{3}{y^{\frac{1}{3}}} = x^3 + C$$

$$-3 = (x^3 + C)y^{\frac{1}{3}}$$

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$$y = \left(-\frac{3}{x^3 + C}\right)^3$$

We'll substitute the initial condition $y(0) = y_0$ to find a value for C.

$$y_0 = \left(-\frac{3}{0^3 + C}\right)^3$$

$$C^3 = -\frac{27}{y_0}$$

$$C = -\frac{3}{\sqrt[3]{y}_0}$$

Now we can plug this value back into the general solution, which will give us the equation's actual solution.

$$y = \left(-\frac{3}{x^3 - \frac{3}{\sqrt[3]{y}_0}}\right)^3$$

$$y = \left(-\frac{3\sqrt[3]{y}_0}{x^3\sqrt[3]{y}_0 - 3}\right)^3$$

$$y = -\frac{27y_0}{(x^3\sqrt[3]{y}_0 - 3)^3}$$

This particular solution is undefined where the denominator is 0.

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$$(x^3\sqrt[3]{y}_0 - 3)^3 = 0$$

$$x^3\sqrt[3]{y}_0 - 3 = 0$$

$$x = \frac{\sqrt[3]{3}}{\sqrt[9]{y}_0}$$

Because the solution is undefined at this value, it divides $(-\infty, \infty)$ into two potential intervals of validity.

$$-\infty < x < \frac{\sqrt[3]{3}}{\sqrt[9]{y}_0} \qquad \frac{\sqrt[3]{3}}{\sqrt[9]{y}_0} < x < \infty$$

Since we're using the initial condition $y(0) = y_0$, and $y_0 > 0$, the interval of validity is

$$\left(-\infty, \frac{\sqrt[3]{3}}{\sqrt[9]{y}_0}\right)$$

■ 5. Find the interval of validity for the solution to the differential equation.

$$\frac{dy}{dx} = y^2e^x$$

$$y(0) = \frac{1}{e^2 - 1}$$

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Solution:

This is a separable differential equation, which means it's non-linear, and therefore we'll need to find the solution to the equation before we can find the interval of validity. We'll first find the general solution by separating variables,

$$\frac{dy}{dx} = y^2 e^x$$

$$dy = y^2 e^x \, dx$$

$$\frac{1}{y^2} \, dy = e^x \, dx$$

integrating both sides, and then solving for y.

$$\int \frac{1}{y^2} \, dy = \int e^x \, dx$$

$$-\frac{1}{y} = e^x + C$$

$$y = -\frac{1}{e^x + C}$$

We'll substitute the initial condition $y(0) = 1/(e^2 - 1)$ to find a value for C.

$$\frac{1}{e^2 - 1} = -\frac{1}{e^0 + C}$$

$$e^2 - 1 = -(e^0 + C)$$

$$e^2 - 1 = -1 - C$$

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$$C = -e^2$$

Now we can plug this value back into the general solution, which will give us the equation's actual solution.

$$y = -\frac{1}{e^x - e^2}$$

The general solution is undefined when

$$e^x - e^2 = 0$$

$$e^x = e^2$$

$$x = 2$$

Because the solution is undefined at this value, it divides $(-\infty, \infty)$ into two potential intervals of validity.

$$-\infty < x < 2 \qquad\qquad 2 < x < \infty$$

Since we're using the initial condition $y(0) = 1/(e^2 - 1)$, and $x_0 = 0$ is contained within $(-\infty,2)$, we can say that $(-\infty,2)$ is the interval of validity.

■ 6. Find the interval of validity for the solution to the differential equation, given $y(1) = y_0$.

$$xy' = y^2$$

Solution:

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This is a separable differential equation, which means it's non-linear, and therefore we'll need to find the solution to the equation before we can find the interval of validity. We'll first find the general solution by separating variables,

$$\frac{dy}{dx} = \frac{y^2}{x}$$

$$dy = \frac{y^2}{x}\, dx$$

$$\frac{1}{y^2}\, dy = \frac{1}{x}\, dx$$

integrating both sides, and then solving for y.

$$\int \frac{1}{y^2}\, dy = \int \frac{1}{x}\, dx$$

$$-\frac{1}{y} = \ln|x| + C$$

$$y = -\frac{1}{\ln|x| + C}$$

We'll substitute the initial condition $y(1) = y_0$ to find a value for C.

$$y_0 = -\frac{1}{\ln|1| + C}$$

$$y_0 = -\frac{1}{C}$$

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$$C = -\frac{1}{y_0}$$

Now we can plug this value back into the general solution, which will give us the equation's actual solution.

$$y = -\frac{1}{\ln|x| - \frac{1}{y_0}}$$

$$y = \frac{-y_0}{\ln|x|y_0 - 1}$$

This particular solution is undefined where the denominator is 0.

$$\ln|x|y_0 - 1 = 0$$

$$\ln|x|y_0 = 1$$

$$\ln|x| = \frac{1}{y_0}$$

$$|x| = e^{\frac{1}{y_0}}$$

$$x = \pm e^{\frac{1}{y_0}}$$

Because we have $\ln|x|$ in the actual solution, we also have to have $x \neq 0$. Which means we've divided $(-\infty, \infty)$ into four potential intervals of validity.

$$-\infty < x < -e^{\frac{1}{y_0}} \qquad -e^{\frac{1}{y_0}} < x < 0 \qquad 0 < x < e^{\frac{1}{y_0}} \qquad e^{\frac{1}{y_0}} < x < \infty$$

Since we're using the initial condition $y(1) = y_0$, the interval of validity must contain $x = 1$. The third interval will only contain $x = 1$ when

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$$e^{\frac{1}{y_0}} > 1$$

$$\ln(e^{\frac{1}{y_0}}) > \ln 1$$

$$\frac{1}{y_0} > 0$$

$$y_0 > 0$$

The fourth interval will only contain $x = 1$ when

$$e^{\frac{1}{y_0}} < 1$$

$$\ln(e^{\frac{1}{y_0}}) < \ln 1$$

$$\frac{1}{y_0} < 0$$

$$y_0 < 0$$

So we've found constraints for $y_0 < 0$ and $y_0 > 0$, but what about $y_0 = 0$? If we substitute $y_0 = 0$ into the actual solution, we get

$$y = \frac{-y_0}{\ln|x|\,y_0 - 1}$$

$$0 = \frac{-0}{\ln|x|\,(0) - 1}$$

This is true for all x, which means the interval of validity is $(-\infty, \infty)$ when $y_0 = 0$. So we can summarize our findings as

For $y_0 < 0$, the interval of validity is $e^{\frac{1}{y_0}} < x < \infty$

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For $y_0 = 0$, the interval of validity is $(-\infty, \infty)$

For $y_0 > 0$, the interval of validity is $0 < x < e^{\frac{1}{y_0}}$

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EULER'S METHOD

■ 1. If $y(0) = 0$, use Euler's method to approximate $y(2)$ with $n = 4$ steps.

$$y' = 1 + y$$

Solution:

We're starting at $y(0)$ and we need to get to $y(2)$ in $n = 4$ steps, so

$$\Delta t = \frac{2 - 0}{4}$$

$$\Delta t = \frac{1}{2}$$

Starting with $(t_0, y_0) = (0,0)$, and with $\Delta t = 1/2$, we build our table and use it to approximate $y(2)$.

$t_0 = 0$	$y_0 = 0$	$y_0 = 0$
$t_1 = \frac{1}{2}$	$y_1 = 0 + (1 + 0)\left(\frac{1}{2}\right)$	$y_1 = \frac{1}{2}$
$t_2 = 1$	$y_2 = \frac{1}{2} + \left(1 + \frac{1}{2}\right)\left(\frac{1}{2}\right)$	$y_2 = \frac{5}{4}$
$t_3 = \frac{3}{2}$	$y_3 = \frac{5}{4} + \left(1 + \frac{5}{4}\right)\left(\frac{1}{2}\right)$	$y_3 = \frac{19}{8}$

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$$t_4 = 2 \qquad y_4 = \frac{19}{8} + \left(1 + \frac{19}{8}\right)\left(\frac{1}{2}\right) \qquad y_4 = \frac{65}{16}$$

$$y_4 \approx 4.06$$

So, using $n = 4$ steps, Euler's method approximates $y(2) \approx 4.06$.

■ 2. If $y(2) = 0$, use Euler's method to approximate $y(4)$ with $n = 5$ steps.

$$y' = 3t + y$$

Solution:

We're starting at $y(2)$ and we need to get to $y(4)$ in $n = 5$ steps, so

$$\Delta t = \frac{4 - 2}{5}$$

$$\Delta t = \frac{2}{5}$$

Starting with $(t_0, y_0) = (2,0)$, and with $\Delta t = 2/5$, we can build our table and use it to approximate $y(4)$.

$$t_0 = 2 \qquad y_0 = 0 \qquad y_0 = 0$$

$$t_1 = \frac{12}{5} \qquad y_1 = 0 + (3(2) + 0)\left(\frac{2}{5}\right) \qquad y_1 = \frac{12}{5}$$

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$$t_2 = \frac{14}{5} \qquad y_2 = \frac{12}{5} + \left(3\left(\frac{12}{5}\right) + \frac{12}{5}\right)\left(\frac{2}{5}\right) \qquad y_2 = \frac{156}{25}$$

$$t_3 = \frac{16}{5} \qquad y_3 = \frac{156}{25} + \left(3\left(\frac{14}{5}\right) + \frac{156}{25}\right)\left(\frac{2}{5}\right) \qquad y_3 = \frac{1{,}512}{125}$$

$$t_4 = \frac{18}{5} \qquad y_4 = \frac{1{,}512}{125} + \left(3\left(\frac{16}{5}\right) + \frac{1{,}512}{125}\right)\left(\frac{2}{5}\right) \qquad y_4 = \frac{12{,}984}{625}$$

$$t_5 = 4 \qquad y_5 = \frac{12{,}984}{625} + \left(3\left(\frac{18}{5}\right) + \frac{12{,}984}{625}\right)\left(\frac{2}{5}\right) \qquad y_5 = \frac{104{,}388}{3{,}125}$$

$$y_5 \approx 33.4$$

So, using $n = 5$ steps, Euler's method approximates $y(4) \approx 33.4$.

■ 3. If $y(0) = 1$, use Euler's method to approximate $y(1)$ with $n = 4$ steps.

$$y' = 1 - 2y$$

Solution:

We're starting at $y(0)$ and we need to get to $y(1)$ in $n = 4$ steps, so

$$\Delta t = \frac{1 - 0}{4}$$

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$$\Delta t = \frac{1}{4}$$

Starting with $(t_0, y_0) = (0,1)$, and with $\Delta t = 1/4$, we build our table and use it to approximate $y(1)$.

$$t_0 = 0 \qquad y_0 = 1 \qquad\qquad y_0 = 1$$

$$t_1 = \frac{1}{4} \qquad y_1 = 1 + (1 - 2(1))\left(\frac{1}{4}\right) \qquad y_1 = \frac{3}{4}$$

$$t_2 = \frac{1}{2} \qquad y_2 = \frac{3}{4} + \left(1 - 2\left(\frac{3}{4}\right)\right)\left(\frac{1}{4}\right) \qquad y_2 = \frac{5}{8}$$

$$t_3 = \frac{3}{4} \qquad y_3 = \frac{5}{8} + \left(1 - 2\left(\frac{5}{8}\right)\right)\left(\frac{1}{4}\right) \qquad y_3 = \frac{9}{16}$$

$$t_4 = 1 \qquad y_4 = \frac{9}{16} + \left(1 - 2\left(\frac{9}{16}\right)\right)\left(\frac{1}{4}\right) \qquad y_4 = \frac{17}{32}$$

$$y_4 \approx 0.53$$

So, using $n = 4$ steps, Euler's method approximates $y(1) \approx 0.53$.

■ 4. Use Euler's method and five steps to approximate $y(3)$, given $y(2) = 1$.

$$y' = t^2 + y$$

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Solution:

We're starting at $y(2)$ and we need to get to $y(3)$ in $n = 5$ steps, so

$$\Delta t = \frac{3-2}{5}$$

$$\Delta t = \frac{1}{5}$$

Starting with $(t_0, y_0) = (2,1)$, and with $\Delta t = 1/5$, we can build our table and use it to approximate $y(3)$.

$$t_0 = 2 \qquad y_0 = 1 \qquad\qquad y_0 = 1$$

$$t_1 = \frac{11}{5} \qquad y_1 = 1 + (2^2 + 1)\left(\frac{1}{5}\right) \qquad y_1 = 2$$

$$t_2 = \frac{12}{5} \qquad y_2 = 2 + \left(\left(\frac{11}{5}\right)^2 + 2\right)\left(\frac{1}{5}\right) \qquad y_2 = \frac{421}{125}$$

$$t_3 = \frac{13}{5} \qquad y_3 = \frac{421}{125} + \left(\left(\frac{12}{5}\right)^2 + \frac{421}{125}\right)\left(\frac{1}{5}\right) \qquad y_3 = \frac{3{,}246}{625}$$

$$t_4 = \frac{14}{5} \qquad y_4 = \frac{3{,}246}{625} + \left(\left(\frac{13}{5}\right)^2 + \frac{3{,}246}{625}\right)\left(\frac{1}{5}\right) \qquad y_4 = \frac{23{,}701}{3{,}125}$$

$$t_5 = 3 \qquad y_5 = \frac{23{,}701}{3{,}125} + \left(\left(\frac{14}{5}\right)^2 + \frac{23{,}701}{3{,}125}\right)\left(\frac{1}{5}\right) \qquad y_5 = \frac{166{,}706}{15{,}625}$$

$$y_5 \approx 10.67$$

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So, using $n = 5$ steps, Euler's method approximates $y(3) \approx 10.67$.

■ 5. If $y(0) = 0$, use Euler's method to approximate $y(1)$ with $\Delta t = 0.2$.

$$y' + 3y = 1 - e^{-5t}$$

Solution:

Starting with $(t_0, y_0) = (0,0)$, and with $\Delta t = 0.2$, we can build our table and use it to approximate $y(1)$.

$t_0 = 0$	$y_0 = 0$	$y_0 = 0$
$t_1 = 0.2$	$y_1 = 0 + (-3(0) + 1 - e^{-5(0)})(0.2)$	$y_1 = 0$
$t_2 = 0.4$	$y_2 = 0 + (-3(0) + 1 - e^{-5(0.2)})(0.2)$	$y_2 \approx 0.1264$
$t_3 = 0.6$	$y_3 = 0.1264 + (-3(0.1264) + 1 - e^{-5(0.4)})(0.2)$	$y_3 \approx 0.2235$
$t_4 = 0.8$	$y_4 = 0.2235 + (-3(0.2235) + 1 - e^{-5(0.6)})(0.2)$	$y_4 \approx 0.2794$
$t_5 = 1$	$y_5 = 0.2794 + (-3(0.2794) + 1 - e^{-5(0.8)})(0.2)$	$y_5 \approx 0.3081$

So, using $n = 5$ steps, Euler's method approximates $y(1) \approx 0.3081$.

■ 6. If $y(0) = 3$, use Euler's method to approximate y_3 with $\Delta t = 0.1$, find the exact value and percentage error.

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$$y' = 8e^{2t} - 3y$$

Solution:

Starting with $(t_0, y_0) = (0,3)$, and with $\Delta t = 0.1$, we can build our table and use it to approximate y_3.

$t_0 = 0$	$y_0 = 3$	$y_0 = 3$
$t_1 = 0.1$	$y_1 = 3 + (8e^{2(0)} - 3(3))(0.1)$	$y_1 = 2.9$
$t_2 = 0.2$	$y_2 = 2.9 + (8e^{2(0.1)} - 3(2.9))(0.1)$	$y_2 \approx 3.0071$
$t_3 = 0.3$	$y_2 = 3.0071 + (8e^{2(0.2)} - 3(3.0071))(0.1)$	$y_3 \approx 3.2984$

So Euler's method approximates $y(0.3) \approx 3.2984$.

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AUTONOMOUS EQUATIONS AND EQUILIBRIUM SOLUTIONS

■ 1. Find any equilibrium solutions of the autonomous differential equation, then determine whether each solution is stable, unstable, or semi-stable without sketching a direction field.

$$\frac{dy}{dt} = -y^2 - y + 6$$

Solution:

The autonomous differential equation has equilibrium solutions at

$$-y^2 - y + 6 = 0$$

$$-(y + 3)(y - 2) = 0$$

$$y = -3, 2$$

These two equilibrium solutions divide the vertical axis into three intervals:

$$y < -3$$

$$-3 < y < 2$$

$$2 < y$$

We need a test-value in each of the three intervals, so we'll use $y = -4$, $y = 0$, and $y = 3$, then substitute these values into $f(y) = -y^2 - y + 6$.

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$$f(-4) = -(-4)^2 - (-4) + 6 = -16 + 4 + 6 = -6 < 0$$

$$f(0) = -(0)^2 - (0) + 6 = 0 - 0 + 6 = 6 > 0$$

$$f(3) = -(3)^2 - (3) + 6 = -9 - 3 + 6 = -6 < 0$$

We find a negative value when $y = -4$, a positive value when $y = 0$, and a negative value when $y = 3$, so we can summarize these results in a table.

Interval	**Sign of** $f(y)$	**Direction of** $f(y)$
$(2,\infty)$	$-$	Decreasing/Falling
$(-3,2)$	$+$	Increasing/Rising
$(-\infty,-3)$	$-$	Decreasing/Falling

The solution curves approach $y = 2$ on both sides, but move away from $y = -3$ on both sides. Therefore, $y = 2$ is a stable equilibrium solution, while $y = -3$ is an unstable equilibrium solution.

■ 2. Find any equilibrium solutions of the autonomous differential equation, then determine whether each solution is stable, unstable, or semi-stable without sketching a direction field.

$$\frac{dy}{dt} = y^3 - 4y^2 + 4y$$

Solution:

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The autonomous differential equation has equilibrium solutions at

$$y^3 - 4y^2 + 4y = 0$$

$$y(y-2)(y-2) = 0$$

$$y = 0,\ 2$$

These two equilibrium solutions divide the vertical axis into three intervals:

$$y < 0$$

$$0 < y < 2$$

$$2 < y$$

We need a test-value in each of the three intervals, so we'll use $y = -1$, $y = 1$, and $y = 3$, then substitute these values into $f(y) = y^3 - 4y^2 + 4y$.

$$f(-1) = (-1)^3 - 4(-1)^2 + 4(-1) = -1 - 4 - 4 = -9 < 0$$

$$f(1) = (1)^3 - 4(1)^2 + 4(1) = 1 - 4 + 4 = 1 > 0$$

$$f(3) = (3)^3 - 4(3)^2 + 4(3) = 27 - 36 + 12 = 3 > 0$$

We find a negative value when $y = -9$, and positive values when $y = 1$ and when $y = 3$, so we can summarize these results in a table.

Interval	**Sign of** $f(y)$	**Direction of** $f(y)$
$(2,\infty)$	$+$	Increasing/Rising
$(0,2)$	$+$	Increasing/Rising

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$(-\infty,0)$	–	Decreasing/Falling

The solution curves move away from $y = 0$ on both sides, so $y = 0$ is an unstable equilibrium solution. The solution curves move toward $y = 2$ on one side and away from it on another side, so $y = 2$ is a semi-stable equilibrium solution.

■ 3. Find any equilibrium solutions of the autonomous differential equation, then determine whether each solution is stable, unstable, or semi-stable.

$$\frac{dy}{dt} = (y^2 - 9)(y - 1)^2$$

Solution:

The autonomous differential equation has equilibrium solutions at

$$(y^2 - 9)(y - 1)^2 = 0$$

$$(y - 3)(y + 3)(y - 1)^2 = 0$$

$$y = -3,\ 1,\ 3$$

These three equilibrium solutions divide the vertical axis into four intervals:

$$y < -3$$

$$-3 < y < 1$$

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$$1 < y < 3$$

$$3 < y$$

We need a test-value in each of the four intervals, so we'll use $y = -4$, $y = 0$, $y = 2$, and $y = 4$, then substitute these values into $f(y) = (y^2 - 9)(y - 1)^2$.

$$f(-4) = ((-4)^2 - 9)(-4 - 1)^2 = (16 - 9)(-5)^2 = 175 > 0$$

$$f(0) = ((0)^2 - 9)(0 - 1)^2 = (0 - 9)(-1)^2 = -9 < 0$$

$$f(2) = (2^2 - 9)(2 - 1)^2 = (4 - 9)(1)^2 = -5 < 0$$

$$f(4) = (4^2 - 9)(4 - 1)^2 = (16 - 9)(3)^2 = 63 > 0$$

We find positive values when $y = -4$ and when $y = 4$, and negative values when $y = 0$ and when $y = 2$, so we can summarize these results in a table.

Interval	**Sign of $f(y)$**	**Direction of $f(y)$**
$(3,\infty)$	$+$	Increasing/Rising
$(1,3)$	$-$	Decreasing/Falling
$(-3,1)$	$-$	Decreasing/Falling
$(-\infty,-3)$	$+$	Increasing/Rising

The solution curves move toward $y = -3$ on both sides, so $y = -3$ is a stable equilibrium solution. The solution curves move away from $y = 3$ on both sides, so $y = 3$ is an unstable equilibrium solution. The solution curves are moving toward $y = 1$ on one side and away from it on another side, so $y = 1$ is a semi-stable equilibrium solution.

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■ 4. Find any equilibrium solutions of the autonomous differential equation, then determine whether each solution is stable, unstable, or semi-stable.

$$\frac{dy}{dt} = \sin y$$

Solution:

The autonomous differential equation has equilibrium solutions at

$$\sin y = 0$$

$y = \pi k$ for any integer k

These equilibrium solutions divide the vertical axis into an infinite number of intervals:

$$\ldots$$

$$-2\pi < y < -\pi$$

$$-\pi < y < 0$$

$$0 < y < \pi$$

$$\pi < y < 2\pi$$

$$\ldots$$

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We need a test-value in each interval, so we'll use $y = -3\pi/2$, $y = -\pi/2$, $y = \pi/2$, and $y = 3\pi/2$, then substitute these values into $f(y) = \sin y$.

...

$$f\left(-\frac{3\pi}{2}\right) = \sin\left(-\frac{3\pi}{2}\right) = 1 > 0$$

$$f\left(-\frac{\pi}{2}\right) = \sin\left(-\frac{\pi}{2}\right) = -1 < 0$$

$$f\left(\frac{\pi}{2}\right) = \sin\left(\frac{\pi}{2}\right) = 1 > 0$$

$$f\left(\frac{3\pi}{2}\right) = \sin\left(\frac{3\pi}{2}\right) = -1 < 0$$

...

We find alternating positive and negative values as we move from one test interval to the next. We'll summarize these results in a table.

Interval	**Sign of** $f(y)$	**Direction of** $f(y)$
...	+	
$(\pi,2\pi)$	−	Decreasing/Falling
$(0,\pi)$	+	Increasing/Rising
$(-\pi,0)$	−	Decreasing/Falling
$(-2\pi,-\pi)$	+	Increasing/Rising

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... –

Based on this pattern, the solution curves will move toward

$$y = \ldots -5\pi,\ -3\pi,\ -\pi,\ \pi,\ 3\pi,\ 5\pi,\ \ldots$$

$y = \pi + 2\pi k$ for any integer k

so these are all stable equilibrium solutions. On the other hand, the solution curves move away from

$$y = \ldots -4\pi,\ -2\pi,\ 0,\ 2\pi,\ 4\pi,\ \ldots$$

$y = 2\pi k$ for any integer k

so these are all unstable equilibrium solutions.

■ 5. Find any equilibrium solutions of the autonomous differential equation, then determine whether each solution is stable, unstable, or semi-stable.

$$\frac{dy}{dt} = |16 - y^2|$$

Solution:

The autonomous differential equation has equilibrium solutions at

$$|16 - y^2| = 0$$

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$$16 - y^2 = 0$$

$$(4 - y)(4 + y) = 0$$

$$y = -4, 4$$

These two equilibrium solutions divide the vertical axis into three intervals:

$$y < -4$$

$$-4 < y < 4$$

$$4 < y$$

We need a test-value in each of the three intervals, so we'll use $y = -5$, $y = 0$, and $y = 5$, then substitute these values into $f(y) = |16 - y^2|$.

$$f(-5) = |16 - (-5)^2| = |16 - 25| = 9 > 0$$

$$f(0) = |16 - 0^2| = |16 - 0| = 16 > 0$$

$$f(5) = |16 - 5^2| = |16 - 25| = 9 > 0$$

We find positive values in each interval, which means both equilibrium solutions are semi-stable.

■ 6. Find any equilibrium solutions of the autonomous differential equation, then determine whether each solution is stable, unstable, or semi-stable.

$$\frac{dy}{dt} + 3y + 6 = e^y(3y + 6)$$

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Solution:

The autonomous differential equation has equilibrium solutions at

$$e^y(3y+6)-3y-6=0$$

$$e^y(3y+6)-(3y+6)=0$$

$$(3y+6)(e^y-1)=0$$

$$y=0,\ -2$$

These two equilibrium solutions divide the vertical axis into three intervals:

$$y<-2$$

$$-2<y<0$$

$$0<y$$

We need a test-value in each of the three intervals, so we'll use $y=-4$, $y=-1$, and $y=1$, then substitute these values into $f(y)=e^y(3y+6)-3y-6$.

$$f(-4)=e^{-4}(3(-4)+6)-3(-4)-6=\frac{1}{e^4}(-6)+6=-6\left(\frac{1-e^4}{e^4}\right)>0$$

$$f(-1)=e^{-1}(3(-1)+6)-3(-1)-6=\frac{1}{e}(3)-3=3\left(\frac{1-e}{e}\right)<0$$

$$f(1)=e^1(3(1)+6)-3(1)-6=e(9)-9=9e-9>0$$

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We find positive values when $y = -4$ and when $y = 1$, and a negative value when $y = -1$, so we can summarize these results in a table.

Interval	**Sign of** $f(y)$	**Direction of** $f(y)$
$(0,\infty)$	$+$	Increasing/Rising
$(-2,0)$	$-$	Decreasing/Falling
$(-\infty,-2)$	$+$	Increasing/Rising

The solution curves move toward $y = -2$ on both sides, so $y = -2$ is a stable equilibrium solution. And the solution curves move away from $y = 0$ on both sides, so $y = 0$ is an unstable equilibrium solution.

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THE LOGISTIC EQUATION

■ 1. Describe the growth pattern of a population of butterflies, assuming their population is modeled by the logistic growth equation, if the carrying capacity of their habitat is $3{,}700$ butterflies.

Solution:

Between $P = 0$ and $P = M = 3{,}700$, the butterfly population is always increasing. In other words, in the interval $0 < P < 3{,}700$, the butterfly population will grow slowly when P is close to 0, then growth will speed up, and then the growth rate will level off again as P gets close to $3{,}700$.

If the population starts at a size of exactly 0, of course it will neither increase nor decrease, as indicated by the equilibrium solution at $P = 0$.

If the population starts at a size of exactly $3{,}700$, then it will neither increase nor decrease, as indicated by the equilibrium solution at $P = 3{,}700$.

And if the butterfly population starts at a size greater than $3{,}700$, it will die off until it reaches stability at a size of $3{,}700$, as indicated by the stable equilibrium solution at $P = 3{,}700$.

■ 2. A bacteria population is observed to be $2{,}500$ at $t = 0$. Using the logistic growth model and a growth rate of 0.03 with a carrying capacity of $4{,}000$, write the logistic growth equation, then solve the initial value problem.

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Solution:

Knowing that $k = 0.03$ and $M = 4{,}000$, we'll substitute into the logistic growth equation.

$$\frac{dP}{dt} = kP\left(1 - \frac{P}{M}\right)$$

$$\frac{dP}{dt} = 0.03P\left(1 - \frac{P}{4{,}000}\right)$$

$$\frac{dP}{dt} = 0.03P\left(\frac{4{,}000 - P}{4{,}000}\right)$$

$$\frac{dP}{dt} = \frac{0.03P(4{,}000 - P)}{4{,}000}$$

$$\frac{dP}{dt} = \frac{0.03}{4{,}000}(4{,}000P - P^2)$$

Separate variables, then integrate both sides of the equation.

$$\frac{1}{4{,}000P - P^2}\,dP = \frac{0.03}{4{,}000}\,dt$$

$$\int \frac{1}{4{,}000P - P^2}\,dP = \int \frac{0.03}{4{,}000}\,dt$$

$$\int \frac{1}{4{,}000P - P^2}\,dP = \frac{0.03}{4{,}000}t + C$$

Apply a partial fractions decomposition.

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$$\frac{1}{P(4{,}000-P)}=\frac{A}{P}+\frac{B}{4{,}000-P}$$

$$1=A(4{,}000-P)+BP$$

$$1=4{,}000A-AP+BP$$

$$1=(-A+B)P+4{,}000A$$

Equating coefficients tells us that $A=1/4{,}000$, then that $-A+B=(-1/4{,}000)+B=0$, and therefore that $B=1/4{,}000$.

$$\int \frac{\frac{1}{4{,}000}}{P}+\frac{\frac{1}{4{,}000}}{4{,}000-P}\,dP=\frac{0.03}{4{,}000}t+C$$

$$\frac{1}{4{,}000}\int \frac{1}{P}+\frac{1}{4{,}000-P}\,dP=\frac{0.03}{4{,}000}t+C$$

$$\frac{1}{4{,}000}(\ln|P|-\ln|4{,}000-P|)=\frac{0.03}{4{,}000}t+C$$

$$\ln|P|-\ln|4{,}000-P|=0.03t+C$$

$$\ln\left|\frac{P}{4{,}000-P}\right|=0.03t+C$$

Solve for P to get the general solution.

$$\left|\frac{P}{4{,}000-P}\right|=e^{0.03t+C}$$

$$\frac{P}{4{,}000-P}=Ce^{0.03t}$$

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Substitute the initial condition $P(0) = 2{,}500$.

$$\frac{2{,}500}{4{,}000 - 2{,}500} = Ce^{0.03(0)}$$

$$\frac{2{,}500}{1{,}500} = C$$

$$C = \frac{5}{3}$$

Then the actual solution is

$$\frac{P}{4{,}000 - P} = \frac{5}{3}e^{0.03t}$$

$$\frac{3P}{4{,}000 - P} = 5e^{0.03t}$$

■ 3. An insect population is observed to be 500 at time $t = 0$. Assuming the population follows a logistic growth model with a growth rate of 0.01 and a carrying capacity of $2{,}500$, solve the initial value problem.

Solution:

Knowing that $k = 0.01$ and $M = 2{,}500$, we'll substitute into the logistic growth equation.

$$\frac{dP}{dt} = kP\left(1 - \frac{P}{M}\right)$$

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$$\frac{dP}{dt} = 0.01P\left(1 - \frac{P}{2{,}500}\right)$$

$$\frac{dP}{dt} = 0.01P\left(\frac{2{,}500 - P}{2{,}500}\right)$$

$$\frac{dP}{dt} = \frac{0.01P(2{,}500 - P)}{2{,}500}$$

Separate variables and integrate both sides of the equation.

$$\frac{1}{P(2{,}500 - P)}\,dP = \frac{0.01}{2{,}500}\,dt$$

$$\int \frac{1}{P(2{,}500 - P)}\,dP = \int \frac{0.01}{2{,}500}\,dt$$

Apply a partial fractions decomposition.

$$\frac{1}{P(2{,}500 - P)} = \frac{A}{P} + \frac{B}{2{,}500 - P}$$

$$1 = A(2{,}500 - P) + BP$$

$$1 = 2{,}500A - AP + BP$$

$$1 = (-A + B)P + 2{,}500A$$

Equating coefficients tells us that $A = 1/2{,}500$, then that $-A + B = (-1/2{,}500) + B = 0$, and therefore that $B = 1/2{,}500$.

$$\int \frac{\frac{1}{2{,}500}}{P} + \frac{\frac{1}{2{,}500}}{2{,}500 - P}\,dP = \int \frac{0.01}{2{,}500}\,dt$$

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$$\frac{1}{2{,}500}\int \frac{1}{P}+\frac{1}{2{,}500-P}\,dP=\frac{0.01}{2{,}500}t+C$$

$$\frac{1}{2{,}500}(\ln|P|-\ln|2{,}500-P|)=\frac{0.01}{2{,}500}t+C$$

$$\ln|P|-\ln|2{,}500-P|=0.01t+C$$

$$\ln\left|\frac{P}{2{,}500-P}\right|=0.01t+C$$

$$e^{\ln\left|\frac{P}{2{,}500-P}\right|}=e^{0.01t+C}$$

$$\frac{P}{2{,}500-P}=Ce^{0.01t}$$

Substitute the initial condition $P(0)=500$ and solve for C.

$$\frac{500}{2{,}500-500}=Ce^{0.01(0)}$$

$$C=\frac{1}{4}$$

So the actual solution is.

$$\frac{P}{2{,}500-P}=\frac{1}{4}e^{0.01t}$$

■ 4. The population of Canada in $2{,}000$ and 2010 was 30.7 and 34 million people, respectively. Predict its population in 2050 and 2060 assuming an exponential model of population growth.

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Solution:

Since our earliest observation is in 2000, we have $P_0 = P(0) = 30.7$, and after 10 years we find $P(10) = 34$. Substitute these values into exponential growth equation,

$$P(t) = P_0 e^{kt}$$

$$34 = 30.7e^{10k}$$

then solve for k.

$$1.11 = e^{10k}$$

$$\ln 1.11 = 10k$$

$$k \approx 0.01$$

To find the population in 2050, we'll use $t = 2050 - 2000 = 50$,

$$P(50) \approx 30.7e^{0.01(50)}$$

$$P(50) \approx 50.62$$

and to find the population in 2060, we'll use $t = 2060 - 2000 = 60$.

$$P(60) \approx 30.7e^{0.01(60)}$$

$$P(60) \approx 55.94$$

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■ 5. A bacteria population increases 8-fold in 5 hours. Assuming exponential growth, how long did it take for the population to increase 5 -fold?

Solution:

We've been told that the population grows to 8 times its original size. If we call its original size P_0, then we can say that it grows to $8P_0$ after 5 hours. Substituting these values into the exponential growth equation gives

$$8P_0 = P_0 e^{k(5)}$$

$$8 = e^{5k}$$

$$\ln 8 = 5k$$

$$k = \frac{\ln 8}{5}$$

Now that we have a value for the growth constant k, we can can figure out how long it took for the population to increase by 5-fold. If we say that P_0 is the original population, and $5P_0$ is five times the original population, then the exponential equation becomes

$$5P_0 = P_0 e^{\frac{\ln 8}{5}t}$$

$$5 = e^{\frac{\ln 8}{5}t}$$

$$\ln 5 = \ln e^{\frac{\ln 8}{5}t}$$

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$$\ln 5 = \frac{\ln 8}{5}t$$

$$t = \frac{5 \ln 5}{\ln 8}$$

$$t \approx 3.87$$

This result tells us that the bacteria population increased 5-fold in about 3.87 hours, or about 3 hours and 52 minutes.

■ 6. The growth rate of a population of bears is modeled by the differential equation, where time t is measured in years. Solve the differential equation with the initial condition $P(0) = 30$.

$$\frac{dP}{dt} = 0.002P(400 - P)$$

Solution:

Separate variables,

$$\frac{dP}{P(400 - P)} = 0.002 \, dt$$

$$\int \frac{dP}{P(400 - P)} = \int 0.002 \, dt$$

then apply a partial fractions decomposition to integrate both sides.

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$$\frac{1}{P(400-P)} = \frac{A}{P} + \frac{B}{400-P}$$

$$1 = A(400-P) + BP$$

$$1 = 400A - AP + BP$$

$$1 = (-A+B)P + 400A$$

Equating coefficients tells us that $A = 1/400$, then that $-A + B = (-1/400) + B = 0$, and therefore that $B = 1/400$.

$$\int \frac{\frac{1}{400}}{P} + \frac{\frac{1}{400}}{400-P}\, dP = \int 0.002\, dt$$

$$\frac{1}{400}(\ln|P| - \ln|400-P|) = 0.002t + C$$

$$\ln\left|\frac{P}{400-P}\right| = 0.8t + C$$

$$e^{\ln\left|\frac{P}{400-P}\right|} = e^{0.8t+C}$$

$$\frac{P}{400-P} = Ce^{0.8t}$$

Substitute the initial condition $P(0) = 30$ and solve for C.

$$\frac{30}{400-30} = Ce^{0.8(0)}$$

$$C = \frac{3}{37}$$

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So the actual solution to the differential equation is

$$\frac{P}{400-P}=\frac{3}{37}e^{0.8t}$$

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PREDATOR-PREY SYSTEMS

■ 1. Is the system cooperative, competitive, or predator-prey? Find the equilibrium solutions of the system.

$$\frac{dx}{dt} = -0.16x + 0.08xy$$

$$\frac{dy}{dt} = 4.5y - 0.9xy$$

Solution:

The mixed xy term $0.08xy$ in the differential equation for dx/dt is positive, while the mixed xy term $-0.9xy$ in the differential equation for dy/dt is negative, which means this is a predator-prey system.

To find equilibrium solutions, we'll factor both equations to get

$$\frac{dx}{dt} = -0.16x + 0.08xy$$

$$\frac{dx}{dt} = 0.08x(-2 + y)$$

$$\frac{dx}{dt} = 0.08x(y - 2)$$

and

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$$\frac{dy}{dt} = 4.5y - 0.9xy$$

$$\frac{dy}{dt} = 0.9y(5 - x)$$

Setting both equations equal to 0 gives

$$0.08x(y - 2) = 0$$

$0.08x = 0$ and $y - 2 = 0$

$x = 0$ and $y = 2$

and

$$0.9y(5 - x) = 0$$

$0.9y = 0$ and $5 - x = 0$

$y = 0$ and $x = 5$

We need to test both solutions from the first equation with both solutions from the second equation. So if we pair $x = 0$ with both $y = 0$ and $y = 2$, and then in addition we pair $x = 5$ with both $y = 0$ and $y = 2$, we get four combinations.

The solutions of the system are therefore

- $(x, y) = (0,0)$, where both populations are 0 and will remain there
- $(x, y) = (0,2)$, where the y population is stable at 2 while the x population is at 0

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- $(x, y) = (5,0)$, where the x population is stable at 5 while the y population is at 0
- $(x, y) = (5,2)$, where the system is balanced

■ 2. Given the system modeling the interaction between populations of foxes and hares, find any equilibrium where the foxes and hares coexist, and give the size of each population.

$$\frac{dH}{dt} = 0.14H - 0.07FH$$

$$\frac{dF}{dt} = -0.65F + 0.05FH$$

Solution:

To find equilibrium solutions, we'll factor both equations to get

$$\frac{dH}{dt} = 0.14H - 0.07FH$$

$$\frac{dH}{dt} = 0.07H(2 - F)$$

and

$$\frac{dF}{dt} = -0.65F + 0.05FH$$

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$$\frac{dF}{dt} = 0.05F(-13 + H)$$

$$\frac{dF}{dt} = 0.05F(H - 13)$$

Setting both equations equal to 0 gives

$$0.07H(2 - F) = 0$$

$0.07H = 0$ and $2 - F = 0$

$H = 0$ and $F = 2$

and

$0.05F(H - 13) = 0$

$0.05F = 0$ and $H - 13 = 0$

$F = 0$ and $H = 13$

We need to test both solutions from the first equation with both solutions from the second equation. So if we pair $H = 0$ with both $F = 0$ and $F = 2$, and then in addition we pair $H = 13$ with both $F = 0$ and $F = 2$, we get four combinations.

The only equilibrium solution where foxes and hares coexist is $(H, F) = (13,2)$.

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■ 3. Determine whether the system is cooperative, competitive, or predator-prey, and find the maximum size of both populations when they are in equilibrium, assuming both populations are measured in millions.

$$\frac{dx}{dt} = 0.02x - 0.4x^2 + 0.005xy$$

$$\frac{dy}{dt} = 0.1y - 1.1y^2 + 0.05xy$$

Solution:

In both equations, the mixed xy has a positive sign, so the system is cooperative. To find the equilibrium solution, we'll start by factoring the equations.

$$\frac{dx}{dt} = 0.02x - 0.4x^2 + 0.005xy$$

$$\frac{dx}{dt} = 0.005x(4 - 80x + y)$$

$$\frac{dx}{dt} = 0.005x(y - 80x + 4)$$

and

$$\frac{dy}{dt} = 0.1y - 1.1y^2 + 0.05xy$$

$$\frac{dy}{dt} = 0.05y(2 - 22y + x)$$

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$$\frac{dy}{dt} = 0.05y(x - 22y + 2)$$

Setting both equations equal to 0 gives

$$0.005x(y - 80x + 4) = 0$$

$0.005x = 0$ and $y - 80x + 4 = 0$

$x = 0$ and $80x - y = 4$

and

$$0.05y(x - 22y + 2) = 0$$

$0.05y = 0$ and $x - 22y + 2 = 0$

$y = 0$ and $22y - x = 2$

The equilibrium solution pairs are then

$x = 0$ and $y = 0$

$x = 0$ and $22y - x = 2$

$80x - y = 4$ and $y = 0$

$80x - y = 4$ and $22y - x = 2$

The first pair is simply $(x, y) = (0,0)$, and the second and third pairs can be given explicitly with a simple substitution as

$x = 0$ and $22y - x = 2$

$x = 0$ and $22y = 2$

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$$x = 0 \text{ and } y = \frac{1}{11}$$

$$(x, y) = \left(0, \frac{1}{11}\right) \approx (0,\ 0.091)$$

and

$$80x - y = 4 \text{ and } y = 0$$

$$80x = 4 \text{ and } y = 0$$

$$x = \frac{1}{20} \text{ and } y = 0$$

$$(x, y) = \left(\frac{1}{20}, 0\right) \approx (0.05,\ 0)$$

For the last pair, we have to solve the system.

$$80x - y = 4 \text{ and } 22y - x = 2$$

$$(x, y) = \left(\frac{90}{1{,}759}, \frac{164}{1{,}759}\right) \approx (0.051,\ 0.093)$$

Therefore, we have the four combinations

$$(x, y) = (0,\ 0)$$

$$(x, y) \approx (0,\ 0.091)$$

$$(x, y) \approx (0.05,\ 0)$$

$$(x, y) \approx (0.051,\ 0.093)$$

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The last pair is our equilibrium solution. Since the populations are measured in millions, the maximum size of population x is $51{,}000$ and the maximum size of population y is $93{,}000$.

■ 4. Determine whether the system is cooperative, competitive, or predator-prey, and find the equilibrium solutions, assuming both populations are measured in millions.

$$\frac{dx}{dt} = 0.004x - 0.2xy$$

$$\frac{dy}{dt} = 0.03y - 0.36y^2 - 0.6xy$$

Solution:

In both equations, the mixed xy has a negative sign, so the system is competitive. To find the equilibrium solution, we'll start by factoring the equations.

$$\frac{dx}{dt} = 0.004x - 0.2xy$$

$$\frac{dx}{dt} = 0.004x(1 - 50y)$$

and

$$\frac{dy}{dt} = 0.03y - 0.36y^2 - 0.6xy$$

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$$\frac{dy}{dt} = 0.03y(1 - 12y - 20x)$$

Setting both equations equal to 0 gives

$$0.004x(1 - 50y) = 0$$

$0.004x = 0$ and $50y = 1$

$x = 0$ and $y = \frac{1}{50}$

and

$$0.03y(1 - 12y - 20x) = 0$$

$0.03y = 0$ and $1 - 12y - 20x = 0$

$y = 0$ and $12y + 20x = 1$

The equilibrium solution pairs are then

$x = 0$ and $y = 0$

$x = 0$ and $y = \frac{1}{50}$

$12y + 20x = 1$ and $y = 0$

$12y + 20x = 1$ and $y = \frac{1}{50}$

The first and second pairs are simply $(x, y) = (0,0)$ and $(x, y) = (0,1/50)$, and the third and fourth pairs can be determined explicitly with a simple substitution.

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$$12y + 20x = 1 \text{ and } y = 0$$

$$20x = 1 \text{ and } y = 0$$

$$x = \frac{1}{20} \text{ and } y = 0$$

$$(x, y) = \left(\frac{1}{20}, 0\right) = (0.05,\ 0)$$

and

$$12y + 20x = 1 \text{ and } y = \frac{1}{50}$$

$$20x = \frac{19}{25} \text{ and } y = \frac{1}{50}$$

$$x = \frac{19}{500} \text{ and } y = \frac{1}{50}$$

$$(x, y) = \left(\frac{19}{500}, \frac{1}{50}\right) = (0.038,\ 0.02)$$

Therefore, we have the four combinations

$$(x, y) = (0,\ 0)$$

$$(x, y) = (0,\ 0.02)$$

$$(x, y) = (0.05,\ 0)$$

$$(x, y) = (0.038,\ 0.02)$$

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The last pair is our equilibrium solution. Since the populations are measured in millions, the maximum size of population x is $38{,}000$ and the maximum size of population y is $20{,}000$.

■ 5. Two species of bacteria are being cultivated in a petri dish. Every day, an amount of sugar is deposited in the petri dish and the species compete for this source of food. Determine whether the system is cooperative, competitive, or predator-prey and find the equilibrium solutions.

$$\frac{dx}{dt} = 0.001x - 0.03x^2 - 0.004xy$$

$$\frac{dy}{dt} = 0.002y - 0.05y^2 - 0.01xy$$

Solution:

In both equations, the mixed xy has a negative sign, so the system is competitive. To find the equilibrium solution, we'll start by factoring the equations.

$$\frac{dx}{dt} = 0.001x - 0.03x^2 - 0.004xy$$

$$\frac{dx}{dt} = 0.001x(1 - 30x - 4y)$$

and

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$$\frac{dy}{dt} = 0.002y - 0.05y^2 - 0.01xy$$

$$\frac{dy}{dt} = 0.002y(1 - 25y - 5x)$$

Setting both equations equal to 0 gives

$$0.001x(1 - 30x - 4y)$$

$0.001x = 0$ and $1 - 30x - 4y = 0$

$x = 0$ and $30x + 4y = 1$

and

$0.002y(1 - 25y - 5x) = 0$

$0.002y = 0$ and $1 - 25y - 5x = 0$

$y = 0$ and $25y + 5x = 1$

The equilibrium solution pairs are then

$x = 0$ and $y = 0$

$x = 0$ and $30x + 4y = 1$

$25y + 5x = 1$ and $y = 0$

$25y + 5x = 1$ and $30x + 4y = 1$

The first pair is simply $(x, y) = (0,0)$, and the second and third pairs can be determined explicitly with a simple substitution.

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$x = 0$ and $30x + 4y = 1$

$x = 0$ and $4y = 1$

$x = 0$ and $y = \frac{1}{4}$

$(x, y) = \left(0, \frac{1}{4}\right) = (0, 0.25)$

and

$25y + 5x = 1$ and $y = 0$

$5x = 1$ and $y = 0$

$x = \frac{1}{5}$ and $y = 0$

$(x, y) = \left(\frac{1}{5}, 0\right) = (0.2, 0)$

For the last pair, we have to solve the system.

$25y + 5x = 1$ and $30x + 4y = 1$

$(x, y) = \left(\frac{21}{730}, \frac{5}{146}\right) \approx (0.029, 0.034)$

Therefore, we have the four combinations

$(x, y) = (0, 0)$

$(x, y) = (0, 0.25)$

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$$(x, y) = (0.2,\ 0)$$

$$(x, y) \approx (0.029,\ 0.034)$$

■ 6. Is the system cooperative, competitive, or predator-prey? Find the equilibrium solutions of the system.

$$\frac{dx}{dt} = 0.48x - 0.16xy$$

$$\frac{dy}{dt} = 1.2y + 7.2xy$$

Solution:

The mixed xy term $-0.16xy$ in the differential equation for dx/dt is negative, while the mixed xy term $7.2xy$ in the differential equation for dy/dt is positive, which means this is a predator-prey system.

To find equilibrium solutions, we'll factor both equations to get

$$\frac{dx}{dt} = 0.48x - 0.16xy$$

$$\frac{dx}{dt} = 0.16x(3 - y)$$

and

$$\frac{dy}{dt} = 1.2y + 7.2xy$$

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$$\frac{dy}{dt} = 1.2y(1 + 6x)$$

Setting both equations equal to 0 gives

$$0.16x(3 - y) = 0$$

$0.16x = 0$ and $3 - y = 0$

$x = 0$ and $y = 3$

and

$1.2y(1 + 6x) = 0$

$1.2y = 0$ and $1 + 6x = 0$

$y = 0$ and $x = -\frac{1}{6}$

If we pair $x = 0$ with $y = 0$, and we pair $x = -1/6$ with $y = 3$, we get two combinations.

- $(x, y) = (0,0)$, where both populations are 0 and will remain there
- $(x, y) = (-1/6,3)$, where the system is balanced

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EXPONENTIAL GROWTH AND DECAY

■ 1. If world population was 2.56 million in 1950 and 3.04 million in 1960 and is growing exponentially, find a function modeling population in the second half of the 20th century, then use the function to estimate world population in 1984.

Solution:

The problem gives us the initial conditions $P(0) = 2{,}560$ and $P(10) = 3{,}040$. Substituting the first condition into the exponential growth equation, we get

$$P = Ce^{kt}$$

$$2{,}560 = Ce^{k(0)}$$

$$C = P_0 = 2{,}560$$

Now we'll substitute $C = P_0 = 2{,}560$ and $P(10) = 3{,}040$ into the exponential growth equation to find k, the constant of proportionality.

$$3{,}040 = 2{,}560e^{10k}$$

$$\frac{19}{16} = e^{10k}$$

Apply the natural log to both sides.

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$$\ln \frac{19}{16} = \ln(e^{10k})$$

$$\ln \frac{19}{16} = 10k$$

$$k = \frac{1}{10} \ln \frac{19}{16}$$

Therefore, the function modeling population growth is

$$P(t) = 2{,}560e^{\frac{t}{10} \ln \frac{19}{16}}$$

To approximate world population in 1984, we evaluate this function at $t = 1984 - 1950 = 34$.

$$P(34) = 2{,}560e^{\frac{34}{10} \ln \frac{19}{16}}$$

$$P(34) \approx 4{,}591.92$$

Therefore, world population in 1984 was approximately 4.592 million.

■ 2. The size of a population of fleas increased by 4 fold in 2 months. How long did it take for the population to increase 5 fold?

Solution:

The problem gives us the conditions $P(0) = P_0$ and $P(2) = 4P_0$. Substituting the first condition into the exponential growth equation, we get

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$$P = Ce^{kt}$$

$$P_0 = Ce^{k(0)}$$

$$P_0 = C$$

Now we'll substitute $C = P_0$ and $P(2) = 4P_0$ into the exponential growth equation to find k, the constant of proportionality.

$$P = P_0e^{kt}$$

$$4P_0 = P_0e^{k(2)}$$

$$4 = e^{2k}$$

Apply the natural log to both sides.

$$\ln 4 = \ln(e^{2k})$$

$$\ln 4 = 2k$$

$$k = \frac{\ln 4}{2}$$

$$k = \ln 2$$

Now we can use k to find the time it took for the population to increase 5 fold, or the time it took the population to reach a size of $5P_0$.

$$5P_0 = P_0e^{t \ln 2}$$

$$5 = e^{t \ln 2}$$

$$\ln 5 = \ln e^{t \ln 2}$$

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$$\ln 5 = t \ln 2$$

$$t = \frac{\ln 5}{\ln 2}$$

$$t \approx 2.32$$

Therefore, the flea population increased 5 fold in about 2.32 months, or about 2 months and 10 days.

■ 3. The half-life of radium-226 is $1{,}590$ years. If the substance decays exponentially, find a function that models the radioactive decay of a sample of 100 g of radium-226 over time t years. How much of the mass remains after $2{,}000$ years?

Solution:

Since radium-226 decays exponentially, we can write

$$m(t) = m_0 e^{kt}$$

where $m(t)$ is the mass of radium- 226 after t years, m_0 is the initial amount of radium in the sample, and k is the decay constant. From the question we know

$$m(0) = m_0 = 100 \text{ g}$$

$$m(1{,}590) = \frac{m_0}{2} = 50 \text{ g}$$

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We can substitute these values into the exponential decay equation in order to find the decay constant k.

$$m(1{,}590) = m_0 e^{1{,}590k}$$

$$50 = 100e^{1{,}590k}$$

$$\frac{1}{2} = e^{1{,}590k}$$

Apply the natural log to both sides.

$$\ln \frac{1}{2} = \ln(e^{1{,}590k})$$

$$\ln \frac{1}{2} = 1{,}590k$$

$$k = -\frac{1}{1{,}590} \ln 2$$

So the function modeling the decay of radium-226 is

$$m(t) = 100e^{-\frac{t}{1{,}590} \ln 2}$$

We'll substitute $t = 2{,}000$ to determine how much radium-226 remains after $2{,}000$ years.

$$m(2{,}000) = 100e^{-\frac{2{,}000}{1{,}590} \ln 2}$$

$$m(2{,}000) \approx 41.8 \text{ g}$$

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■ 4. A population of $1{,}000$ bacteria, growing exponentially, takes 12 minutes to double in size. How long will it take for the population to reach 1 million?

Solution:

We know $P(0) = P_0 = 1{,}000$ and $P(12) = 2P_0 = 2{,}000$. Substituting the first condition into the exponential growth equation, we get

$$P(t) = P_0 e^{kt}$$

$$2{,}000 = 1{,}000e^{k(12)}$$

$$2 = e^{12k}$$

Apply the natural log to both sides.

$$\ln 2 = \ln(e^{12k})$$

$$\ln 2 = 12k$$

$$k = \frac{1}{12} \ln 2$$

So the growth of the bacteria population can be modeled by

$$P(t) = 1{,}000e^{\frac{t}{12} \ln 2}$$

We'll substitute $P(t) = 1{,}000{,}000$ to determine how long it will take for the population to reach that size.

$$1{,}000{,}000 = 1{,}000e^{\frac{t}{12} \ln 2}$$

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$$1{,}000 = e^{\frac{t}{12}\ln 2}$$

Apply the natural log to both sides.

$$\ln 1{,}000 = \ln e^{\frac{t}{12}\ln 2}$$

$$\ln 1{,}000 = \frac{t}{12}\ln 2$$

$$t = 12\frac{\ln 1{,}000}{\ln 2}$$

$$t \approx 119.6$$

The size of the bacteria population will reach 1 million in approximately 119.6 minutes, or just under two hours.

■ 5. A sofa's price drops by $50\,\%$ in 10 months. If the rate at which the price decreases is proportional to current price, how much will the price decrease in 2 years?

Solution:

We know $P(0) = P_0$ and $P(10) = (1/2)P_0$. Substituting the first condition into the exponential decay equation gives

$$P = Ce^{kt}$$

$$P_0 = Ce^{k(0)}$$

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$$P_0 = C$$

Now we'll substitute $C = P_0$ and $P(10) = (1/2)P_0$ into the exponential equation to find k, the constant of proportionality.

$$P = P_0 e^{kt}$$

$$\frac{1}{2}P_0 = P_0 e^{k(10)}$$

$$\frac{1}{2} = e^{10k}$$

Apply the natural log to both sides.

$$\ln \frac{1}{2} = \ln(e^{10k})$$

$$\ln \frac{1}{2} = 10k$$

$$k = \frac{1}{10} \ln \frac{1}{2}$$

$$k = \frac{1}{10}(\ln 1 - \ln 2)$$

$$k = -\frac{1}{10} \ln 2$$

Now we can use k to find the price after 2 years, or 24 months.

$$P(24) = P_0 e^{\left(-\frac{1}{10} \ln 2\right)24}$$

$$P(24) = P_0 e^{-\frac{12}{5} \ln 2}$$

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$$P(24) = P_0 e^{\ln 2^{-\frac{12}{5}}}$$

$$P(24) = P_0 2^{-\frac{12}{5}}$$

$$P(24) = 0.19P_0$$

The price will fall to $19\,\%$ of the original amount, which means the price will drop by $100\% - 19\,\% = 81\,\%$ after 2 years.

■ 6. A radioactive element decays into nonradioactive substances. After 3 years, the radioactivity decreases by $5\,\%$. How many years would it take for the radioactivity to decrease by $90\,\%$?

Solution:

If we substitute $P(0) = P_0$ into the exponential decay equation, we get

$$P = Ce^{kt}$$

$$P_0 = Ce^{k(0)}$$

$$P_0 = C$$

Now we'll substitute $C = P_0$ and $P(3) = 0.95P_0$, since after 3 years the radioactivity decreases by $5\,\%$, into the decay equation to find k, the constant of proportionality.

$$P = P_0 e^{kt}$$

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$$0.95P_0 = P_0e^{k(3)}$$

$$0.95 = e^{3k}$$

Apply the natural log to both sides.

$$\ln 0.95 = \ln(e^{3k})$$

$$\ln 0.95 = 3k$$

$$k = \frac{1}{3}\ln 0.95$$

Now we can use k and $P(t) = 0.1P_0$ to find the time it takes for radioactivity to decrease by $90\,\%$.

$$0.1P_0 = P_0e^{\frac{t}{3}\ln 0.95}$$

$$0.1 = e^{\frac{t}{3}\ln 0.95}$$

$$\ln 0.1 = \ln e^{\frac{t}{3}\ln 0.95}$$

$$\ln 0.1 = \frac{t}{3}\ln 0.95$$

$$t = \frac{3\ln 0.1}{\ln 0.95}$$

$$t \approx 134.67$$

Therefore, it takes almost 135 years for radioactivity to decrease by $90\,\%$.

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MIXING PROBLEMS

■ 1. A tank contains $5{,}000$ L of water and 100 kg of dissolved salt. Fresh water is entering the tank at 20 L/min, and the solution drains at a rate of 15 L/min. Assuming the solution in the tank remains perfectly mixed, what is the function that models the amount of salt $y(t)$ at any given minute t? How much salt remains in the tank after 15 days?

Solution:

From the question, we know

- $C_1 = 0$ kg/L because the freshwater being added to the tank contains no salt
- $r_1 = 20$ L/min
- $C_2 = \dfrac{y(t) \text{ kg}}{(5{,}000 + 5t) \text{ L}}$ because every minute 20 L enter the tank while 15 L exit, which means 5 L of solution is added to the initial $5{,}000$ L every minute
- $r_2 = 15$ L/min

So the differential equation modeling the change in salt over time is

$$\frac{dy}{dt} = \left(\frac{0 \text{ kg}}{\text{L}}\right)\left(\frac{20 \text{ L}}{\text{min}}\right) - \left(\frac{y(t) \text{ kg}}{(5{,}000 + 5t) \text{ L}}\right)\left(\frac{15 \text{ L}}{\text{min}}\right)$$

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$$\frac{dy}{dt} = \frac{0 \text{ kg}}{\text{min}} - \frac{15y(t) \text{ kg}}{(5{,}000 + 5t) \text{ min}}$$

$$\frac{dy}{dt} = -\frac{3y(t) \text{ kg}}{(1{,}000 + t) \text{ min}}$$

Separate variables and integrate both sides.

$$\int \frac{1}{y}\, dy = \int -\frac{3}{1{,}000 + t}\, dt$$

$$\ln|y| = -3\ln|1{,}000 + t| + C$$

Simplify the equation and solve it for y.

$$e^{\ln|y|} = e^{-3\ln|1{,}000+t|+C}$$

$$|y| = Ce^{-3\ln|1{,}000+t|}$$

$$y = Ce^{\ln|1{,}000+t|^{-3}}$$

$$y = C|1{,}000 + t|^{-3}$$

Then the general solution is

$$y(t) = \frac{C}{(1{,}000 + t)^3}$$

We were told that the tank initially contained 100 kg of dissolved salt, which means $y(0) = 100$ is an initial condition that we can substitute into the general solution to solve for the constant of integration,

$$100 = \frac{C}{(1{,}000 + 0)^3}$$

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$$C = 10^{11}$$

and then we can put this value back into the general solution.

$$y(t) = \frac{10^{11}}{(10^3 + t)^3}$$

To find how much salt remains in the tank after 15 days we just plug in $t = 15(24)(60) = 21{,}600$ minutes to find the amount of salt in the tank after 15 days.

$$y(21{,}600) = \frac{10^{11}}{(1{,}000 + 21{,}600)^3}$$

$$y(21600) = \frac{10^{11}}{22{,}600^3}$$

$$y(21{,}600) \approx 0.008 \approx 8 \times 10^{-3} \text{ kg}$$

■ 2. A tank contains $2{,}000$ L of water and 500 kg of dissolved salt. Fresh water is entering the tank at 20 L/min, and the solution drains at a rate of 50 L/min. Assuming the solution in the tank remains perfectly mixed, what is the function $y(t)$ that models the amount of salt in the tank after t minutes?

Solution:

From the question, we know

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- $C_1 = 0$ kg/L because the freshwater being added to the tank contains no salt
- $r_1 = 20$ L/min
- $C_2 = \dfrac{y(t) \text{ kg}}{(2{,}000 - 30t) \text{ L}}$ because every minute 20 L enter the tank while 50 L exit, which means 30 L of solution is taken away from the initial $2{,}000$ L every minute
- $r_2 = 50$ L/min

So the differential equation modeling the change in salt over time is

$$\frac{dy}{dt} = \left(\frac{0 \text{ kg}}{\text{L}}\right)\left(\frac{20 \text{ L}}{\text{min}}\right) - \left(\frac{y(t) \text{ kg}}{(2{,}000 - 30t) \text{ L}}\right)\left(\frac{50 \text{ L}}{\text{min}}\right)$$

$$\frac{dy}{dt} = \frac{0 \text{ kg}}{\text{min}} - \frac{50y(t) \text{ kg}}{(2{,}000 - 30t) \text{ min}}$$

$$\frac{dy}{dt} = -\frac{5y(t) \text{ kg}}{(200 - 3t) \text{ min}}$$

Separate variables and integrate both sides.

$$\int \frac{1}{y}\, dy = \int -\frac{5}{200 - 3t}\, dt$$

$$\ln|y| = \frac{5}{3}\ln|200 - 3t| + C$$

Simplify the equation and solve it for y.

$$e^{\ln|y|} = e^{\frac{5}{3}\ln|200-3t|+C}$$

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$$|y| = Ce^{\frac{5}{3}\ln|200-3t|}$$

$$y = Ce^{\ln|200-3t|^{\frac{5}{3}}}$$

$$y = C|200 - 3t|^{\frac{5}{3}}$$

We were told that the tank initially contained 500 kg of dissolved salt, which means $y(0) = 500$ is an initial condition that we can substitute into the general solution to solve for the constant of integration,

$$500 = C|200 - 3(0)|^{\frac{5}{3}}$$

$$C \approx 0.073$$

and then we can put this value back into the general solution.

$$y(t) = 0.073|200 - 3t|^{\frac{5}{3}}$$

■ 3. A tank contains $5{,}000$ L of freshwater and is being filled with a brine mixture with a concentration of 150 g/L, at a rate of 20 L/min. Meanwhile, solution drains from the tank at a rate of 30 L/min. Assuming the solution in the tank remains perfectly mixed, what is the function $y(t)$ that models the amount of salt in the tank after t minutes?

Solution:

From the question, we know

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- $C_1 = 0.15$ kg/L because the brine solution being added to the tank has a salt concentration of 150 g/L
- $r_1 = 20$ L/min
- $C_2 = \dfrac{y(t) \text{ kg}}{(5{,}000 - 10t) \text{ L}}$ because every minute 20 L enter the tank while 30 L exit, which means 10 L of solution is taken away from the initial $5{,}000$ L every minute
- $r_2 = 30$ L/min

So the differential equation modeling the change in salt over time is

$$\frac{dy}{dt} = \left(\frac{0.15 \text{ kg}}{\text{L}}\right)\left(\frac{20 \text{ L}}{\text{min}}\right) - \left(\frac{y(t) \text{ kg}}{(5{,}000 - 10t) \text{ L}}\right)\left(\frac{30 \text{ L}}{\text{min}}\right)$$

$$\frac{dy}{dt} = \frac{3 \text{ kg}}{\text{min}} - \frac{30y(t) \text{ kg}}{(5{,}000 - 10t) \text{ min}}$$

$$\frac{dy}{dt} = \frac{3 \text{ kg}}{\text{min}} - \frac{3y(t) \text{ kg}}{(500 - t) \text{ min}}$$

Rewrite the equation in standard form of a linear equation.

$$y' + \left(\frac{3}{500 - t}\right) y = 3$$

Then the integrating factor is

$$I(t) = e^{\int P(t)\, dt}$$

$$I(t) = e^{\int \frac{3}{500 - t}\, dt}$$

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$$I(t) = e^{-3\ln(500-t)}$$

$$I(t) = e^{\ln(500-t)^{-3}}$$

$$I(t) = \frac{1}{(500-t)^3}$$

Then we can rewrite the linear differential equation as

$$\frac{d}{dt}\left(\frac{y}{(500-t)^3}\right) = \frac{3}{(500-t)^3}$$

$$\int \frac{d}{dt}\left(\frac{y}{(500-t)^3}\right)\, dt = \int \frac{3}{(500-t)^3}\, dt$$

$$\frac{y}{(500-t)^3} = \frac{3}{2(500-t)^2} + C$$

So the general solution will be

$$y(t) = \frac{3}{2}(500-t) + C(500-t)^3$$

$$y(t) = 750 - \frac{3}{2}t + C(500-t)^3$$

The problem states that the tank was initially filled with freshwater, which means $y(0) = 0$.

$$0 = 750 - \frac{3}{2}(0) + C(500-0)^3$$

$$C = -\frac{750}{500^3}$$

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$$C = -\frac{3}{500{,}000}$$

So the actual solution is

$$y(t) = 750 - \frac{3}{2}t - \frac{3(500 - t)^3}{500{,}000}$$

■ 4. A tank contains $1{,}000$ L of fresh water. A mix of water and salt is being added into the tank at a rate of 500 L/h, and the solution drains out at the same rate. The concentration of the solution that's being deposited varies periodically, according to the function $f(t) = 2\sin t$ kg/L. Assuming that the solution in the tank remains perfectly mixed, how much salt is in the tank after t hours?

Solution:

From the question, we know

- $C_1 = 2\sin t$ kg/L because the brine solution being added to the tank has a salt concentration modeled by $2\sin t$ kg/L
- $r_1 = 500$ L/h
- $C_2 = \dfrac{y(t) \text{ kg}}{1{,}000 \text{ L}}$ because with equal input and output rates, the amount of solution in the tank stays constant over time
- $r_2 = 500$ L/h

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So the differential equation modeling the change in salt over time is

$$\frac{dy}{dt} = \left(\frac{2 \sin t \text{ kg}}{\text{L}}\right)\left(\frac{500 \text{ L}}{\text{h}}\right) - \left(\frac{y(t) \text{ kg}}{1000 \text{ L}}\right)\left(\frac{500 \text{ L}}{\text{h}}\right)$$

$$\frac{dy}{dt} = \frac{1{,}000 \sin t \text{ kg}}{\text{h}} - \frac{y(t) \text{ kg}}{2 \text{ h}}$$

Rewrite the equation in standard form of a linear equation.

$$y' + \frac{1}{2}y = 1{,}000 \sin t$$

Then the integrating factor is

$$I(t) = e^{\int P(t)\, dt}$$

$$I(t) = e^{\int \frac{1}{2}\, dt}$$

$$I(t) = e^{\frac{t}{2}}$$

Then we can rewrite the linear differential equation as

$$\frac{d}{dt}(ye^{\frac{t}{2}}) = 1{,}000e^{\frac{t}{2}} \sin t$$

$$\int \frac{d}{dt}(ye^{\frac{t}{2}})\, dt = \int 1{,}000e^{\frac{t}{2}} \sin t\, dt$$

$$ye^{\frac{t}{2}} = 1{,}000 \int e^{\frac{t}{2}} \sin t\, dt$$

Use integration by parts with $u = \sin t$, $du = \cos t\, dt$, $dv = e^{\frac{t}{2}}\, dt$, and $v = 2e^{\frac{t}{2}}$ to integrate the right side.

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$$\int e^{\frac{t}{2}} \sin t \ dt = 2e^{\frac{t}{2}} \sin t - 2 \int e^{\frac{t}{2}} \cos t \ dt$$

Use integration by parts again, this time with $u = \cos t$, $du = -\sin t \ dt$, $dv = e^{\frac{t}{2}} \ dt$, and $v = 2e^{\frac{t}{2}}$ to integrate the right side.

$$\int e^{\frac{t}{2}} \sin t \ dt = 2e^{\frac{t}{2}} \sin t - 2 \left[2e^{\frac{t}{2}} \cos t + 2 \int e^{\frac{t}{2}} \sin t \ dt \right]$$

$$\int e^{\frac{t}{2}} \sin t \ dt = 2e^{\frac{t}{2}} \sin t - 4e^{\frac{t}{2}} \cos t - 4 \int e^{\frac{t}{2}} \sin t \ dt$$

$$5 \int e^{\frac{t}{2}} \sin t \ dt = 2e^{\frac{t}{2}} (\sin t - 2 \cos t)$$

$$\int e^{\frac{t}{2}} \sin t \ dt = \frac{2}{5} e^{\frac{t}{2}} (\sin t - 2 \cos t)$$

Substituting back in our equation, we get

$$ye^{\frac{t}{2}} = 1{,}000 \left(\frac{2}{5} e^{\frac{t}{2}} (\sin t - 2 \cos t) \right) + C$$

$$ye^{\frac{t}{2}} = 400e^{\frac{t}{2}} (\sin t - 2 \cos t) + C$$

$$y(t) = 400(\sin t - 2 \cos t) + Ce^{-\frac{t}{2}}$$

Since the tank was filled with fresh water, the initial amount of salt is given by $y(0) = 0$, so we get

$$0 = 400(\sin 0 - 2 \cos 0) + Ce^{-\frac{0}{2}}$$

$$0 = 400(0 - 2) + C$$

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$$0 = -800 + C$$

$$C = 800$$

Plug this value back into the general solution to get the actual solution.

$$y(t) = 400(\sin t - 2\cos t) + 800e^{-\frac{t}{2}}$$

$$y(t) = 400\sin t - 800\cos t + 800e^{-\frac{t}{2}}$$

■ 5. A tank holding $2{,}000$ L of fresh water is being filled at 400 L/h with a brine solution with a concentration of $f(t) = 5 + \cos(t/5)$ kg/L. Assuming that the solution in the tank remains perfectly mixed and drains at a rate equal to the fill rate, how much salt is in the tank after t hours?

Solution:

From the question, we know

- $C_1 = 5 + \cos\left(\frac{t}{5}\right)$ kg/L because the brine solution being added to the tank has a salt concentration modeled by $5 + \cos(t/5)$ kg/L
- $r_1 = 400$ L/h
- $C_2 = \frac{y(t) \text{ kg}}{2{,}000 \text{ L}}$ because with equal input and output rates, the amount of solution in the tank stays constant over time

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- $r_2 = 400$ L/h

So the differential equation modeling the change in salt over time is

$$\frac{dy}{dt} = \left(5 + \cos\left(\frac{t}{5}\right)\right)\frac{\text{kg}}{\text{L}}\left(\frac{400\ \text{L}}{\text{h}}\right) - \left(\frac{y(t)\ \text{kg}}{2{,}000\ \text{L}}\right)\left(\frac{400\ \text{L}}{\text{h}}\right)$$

$$\frac{dy}{dt} = \left(2{,}000 + 400\cos\left(\frac{t}{5}\right)\right)\frac{\text{kg}}{\text{h}} - \frac{y(t)\ \text{kg}}{5\text{h}}$$

Rewrite the equation in standard form of a linear equation.

$$y' + \frac{1}{5}y = 2{,}000 + 400\cos\left(\frac{t}{5}\right)$$

Then the integrating factor is

$$I(t) = e^{\int P(t)\ dt}$$

$$I(t) = e^{\int \frac{1}{5}\ dt}$$

$$I(t) = e^{\frac{t}{5}}$$

Then we can rewrite the linear differential equation as

$$\frac{d}{dt}(ye^{\frac{t}{5}}) = 2{,}000e^{\frac{t}{5}} + 400e^{\frac{t}{5}}\cos\left(\frac{t}{5}\right)$$

$$\int \frac{d}{dt}(ye^{\frac{t}{5}})\ dt = \int 2{,}000e^{\frac{t}{5}} + 400e^{\frac{t}{5}}\cos\left(\frac{t}{5}\right)\ dt$$

$$ye^{\frac{t}{5}} = 2{,}000\int e^{\frac{t}{5}}\ dt + 400\int e^{\frac{t}{5}}\cos\left(\frac{t}{5}\right)\ dt$$

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$$ye^{\frac{t}{5}} = 10{,}000e^{\frac{t}{5}} + 400\int e^{\frac{t}{5}}\cos\left(\frac{t}{5}\right)\,dt$$

Use integration by parts with $u = \cos(t/5)$, $du = -(1/5)\sin(t/5)\,dt$, $dv = e^{\frac{t}{5}}\,dt$, and $v = 5e^{\frac{t}{5}}$ to integrate the right side.

$$\int e^{\frac{t}{5}}\cos\left(\frac{t}{5}\right)\,dt = 5e^{\frac{t}{5}}\cos\left(\frac{t}{5}\right) + \int e^{\frac{t}{5}}\sin\left(\frac{t}{5}\right)\,dt$$

Use integration by parts again, this time with $u = \sin(t/5)$, $du = (1/5)\cos(t/5)\,dt$, $dv = e^{\frac{t}{5}}\,dt$, and $v = 5e^{\frac{t}{5}}$ to integrate the right side.

$$\int e^{\frac{t}{5}}\cos\left(\frac{t}{5}\right)\,dt = 5e^{\frac{t}{5}}\cos\left(\frac{t}{5}\right) + \left[5e^{\frac{t}{5}}\sin\left(\frac{t}{5}\right) - \int e^{\frac{t}{5}}\cos\left(\frac{t}{5}\right)\,dt\right]$$

$$2\int e^{\frac{t}{5}}\cos\left(\frac{t}{5}\right)dt = 5e^{\frac{t}{5}}\cos\left(\frac{t}{5}\right) + 5e^{\frac{t}{5}}\sin\left(\frac{t}{5}\right)$$

$$\int e^{\frac{t}{5}}\cos\left(\frac{t}{5}\right)dt = \frac{5}{2}e^{\frac{t}{5}}\cos\left(\frac{t}{5}\right) + \frac{5}{2}e^{\frac{t}{5}}\sin\left(\frac{t}{5}\right)$$

$$\int e^{\frac{t}{5}}\cos\left(\frac{t}{5}\right)dt = \frac{5}{2}e^{\frac{t}{5}}\left(\cos\left(\frac{t}{5}\right) + \sin\left(\frac{t}{5}\right)\right)$$

Substituting back in our equation, we get

$$ye^{\frac{t}{5}} = 10{,}000e^{\frac{t}{5}} + 400\left[\frac{5}{2}e^{\frac{t}{5}}\left(\cos\left(\frac{t}{5}\right) + \sin\left(\frac{t}{5}\right)\right)\right] + C$$

$$ye^{\frac{t}{5}} = 10{,}000e^{\frac{t}{5}} + 1{,}000e^{\frac{t}{5}}\left(\cos\left(\frac{t}{5}\right) + \sin\left(\frac{t}{5}\right)\right) + C$$

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$$y = 10{,}000 + 1{,}000\left(\cos\left(\frac{t}{5}\right) + \sin\left(\frac{t}{5}\right)\right) + Ce^{-\frac{t}{5}}$$

Since the tank was filled with fresh water, the initial amount of salt is given by $y(0) = 0$, so we get

$$0 = 10{,}000 + 1{,}000\left(\cos\left(\frac{0}{5}\right) + \sin\left(\frac{0}{5}\right)\right) + Ce^{-\frac{0}{5}}$$

$$0 = 10{,}000 + 1{,}000 + C$$

$$C = -11{,}000$$

Plug this value back into the general solution to get the actual solution.

$$y = 10{,}000 + 1{,}000\left(\cos\left(\frac{t}{5}\right) + \sin\left(\frac{t}{5}\right)\right) - 11{,}000e^{-\frac{t}{5}}$$

$$y = 1{,}000\left[\sin\left(\frac{t}{5}\right) + \cos\left(\frac{t}{5}\right) - 11e^{-\frac{t}{5}} + 10\right]$$

■ 6. A tank contains $3{,}000$ L of water and 600 kg of dissolved salt. Fresh water is entering the tank at 10 L/min, and the solution drains out at the same rate. Assuming the solution in the tank remains perfectly mixed, what is the function that models the amount of salt $y(t)$ at any given minute t?

Solution:

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From the question, we know

- $C_1 = 0$ kg/L because the freshwater being added to the tank contains no salt
- $r_1 = 10$ L/min
- $C_2 = \dfrac{y(t) \text{ kg}}{3{,}000 \text{ L}}$ because with equal input and output rates, the amount of solution in the tank stays constant over time
- $r_2 = 10$ L/min

So the differential equation modeling the change in salt over time is

$$\frac{dy}{dt} = \left(\frac{0 \text{ kg}}{\text{L}}\right)\left(\frac{10 \text{ L}}{\text{min}}\right) - \left(\frac{y(t) \text{ kg}}{3{,}000 \text{ L}}\right)\left(\frac{10 \text{ L}}{\text{min}}\right)$$

$$\frac{dy}{dt} = -\frac{y(t) \text{ kg}}{300 \text{ min}}$$

Separate variables and integrate both sides.

$$\int \frac{1}{y}\, dy = \int -\frac{1}{300}\, dt$$

$$\ln|y| = -\frac{1}{300}t + C$$

$$e^{\ln|y|} = e^{-\frac{1}{300}t+C}$$

$$|y| = Ce^{-\frac{1}{300}t}$$

Then the general solution is

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$$y(t) = Ce^{-\frac{1}{300}t}$$

Since the tank initially contained 600 kg of dissolved salt, we know $y(0) = 600$ is an initial condition that we can substitute into the general solution to solve for the constant of integration,

$$600 = Ce^{-\frac{1}{300}(0)}$$

$$C = 600$$

and then we can put this value back into the general solution.

$$y(t) = 600e^{-\frac{1}{300}t}$$

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NEWTON'S LAW OF COOLING

■ 1. The temperature of a soup dropped from $100°C$ to $95°C$ in 5 minutes. How long would it take for the temperature to drop to $40°C$ if ambient temperature is $25°C$?

Solution:

If we plug everything we know,

$T_0 = 100°$ Initial temperature of the soup

$T_a = 25°$ Temperature of the environment

$T(5) = 95°$ At $t = 5$ minutes, the soup has cooled to $95°$

into the Newton's Law of Cooling solution equation, we get

$$T(t) = T_a + (T_0 - T_a)e^{-kt}$$

$$T(t) = 25 + (100 - 25)e^{-kt}$$

$$T(t) = 25 + 75e^{-kt}$$

Substitute the initial condition $T(5) = 95°$,

$$95 = 25 + 75e^{-k(5)}$$

$$95 = 25 + 75e^{-5k}$$

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$$70 = 75e^{-5k}$$

$$\frac{70}{75} = e^{-5k}$$

$$\ln \frac{14}{15} = -5k$$

$$k = -\frac{1}{5} \ln \frac{14}{15}$$

then plug k into the equation modeling temperature over time.

$$T(t) = 25 + 75e^{-\left(-\frac{1}{5} \ln \frac{14}{15}\right)t}$$

$$T(t) = 25 + 75e^{\left(\frac{1}{5} \ln \frac{14}{15}\right)t}$$

We want to find the time t at which the soup reaches $40°$, so we'll substitute $T(t) = 40°$.

$$40 = 25 + 75e^{\left(\frac{1}{5} \ln \frac{14}{15}\right)t}$$

$$15 = 75e^{\left(\frac{1}{5} \ln \frac{14}{15}\right)t}$$

$$\frac{15}{75} = e^{\left(\frac{1}{5} \ln \frac{14}{15}\right)t}$$

$$\frac{1}{5} = e^{\left(\frac{1}{5} \ln \frac{14}{15}\right)t}$$

$$\ln \frac{1}{5} = \frac{t}{5} \ln \frac{14}{15}$$

$$5 \ln \frac{1}{5} = t \ln \frac{14}{15}$$

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$$t = \frac{5 \ln \frac{1}{5}}{\ln \frac{14}{15}}$$

$$t \approx 116.64$$

The soup will cool from $100°C$ to $40°C$ in about 116.64 minutes.

■ 2. The temperature of a liquid dropped from $350°C$ to $300°C$ in 30 minutes. What temperature would the liquid reach in 3 hours if the temperature of the environment is $30°C$?

Solution:

If we plug everything we know,

$T_0 = 350°$ Initial temperature of the liquid

$T_a = 30°$ Temperature of the environment

$T(1/2) = 300°$ At $t = 1/2$ hour, the liquid has cooled to $300°$

into the Newton's Law of Cooling solution equation, we get

$$T(t) = T_a + (T_0 - T_a)e^{-kt}$$

$$T(t) = 30 + (350 - 30)e^{-kt}$$

$$T(t) = 30 + 320e^{-kt}$$

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Substitute the initial condition $T(1/2) = 300°$,

$$300 = 30 + 320e^{-\frac{1}{2}k}$$

$$270 = 320e^{-\frac{1}{2}k}$$

$$\frac{270}{320} = e^{-\frac{1}{2}k}$$

$$\ln\frac{27}{32} = -\frac{1}{2}k$$

$$k = -2\ln\frac{27}{32}$$

then plug k into the equation modeling temperature over time.

$$T(t) = 30 + 320e^{-\left(-2\ln\frac{27}{32}\right)t}$$

$$T(t) = 30 + 320e^{2t\ln\frac{27}{32}}$$

We want to the temperature decrease over 3 hours, so we'll evaluate at $t = 3$ to find $T(3)$.

$$T(3) = 30 + 320e^{2(3)\ln\frac{27}{32}}$$

$$T(3) = 30 + 320e^{6\ln\frac{27}{32}}$$

$$T(3) = 30 + 320e^{\ln\left(\frac{27}{32}\right)^6}$$

$$T(3) = 30 + 320\left(\frac{27}{32}\right)^6$$

$$T(3) \approx 145.46$$

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In 3 hours, the temperature of the liquid will fall by $350°C - 145°C = 205°C$.

■ 3. The temperature of ice cream in a freezer is $-20°$. After two minutes outside the freezer, where the temperature is $35°$, the temperature of the ice cream increased to $-10°$. What will be the temperature of the ice cream after 10 minutes?

Solution:

If we plug everything we know,

$T_0 = -20°$ Initial temperature of the ice cream

$T_a = 35°$ Temperature of the environment

$T(2) = -10°$ At $t = 2$ minutes, the ice cream has warmed to $-10°$

into the Newton's Law of Cooling solution equation, we get

$$T(t) = T_a + (T_0 - T_a)e^{-kt}$$

$$T(t) = 35 + (-20 - 35)e^{-kt}$$

$$T(t) = 35 - 55e^{-kt}$$

Substitute the initial condition $T(2) = -10°$,

$$-10 = 35 - 55e^{-2k}$$

$$-45 = -55e^{-2k}$$

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$$\frac{9}{11} = e^{-2k}$$

$$\ln \frac{9}{11} = -2k$$

$$k = -\frac{1}{2} \ln \frac{9}{11}$$

then plug k into the equation modeling temperature over time.

$$T(t) = 35 - 55e^{-\left(-\frac{1}{2} \ln \frac{9}{11}\right)t}$$

$$T(t) = 35 - 55e^{\frac{1}{2}t \ln \frac{9}{11}}$$

We want to find the temperature of the ice cream after 10 minutes, so we'll evaluate at $t = 10$ to find $T(10)$.

$$T(10) = 35 - 55e^{\frac{1}{2}(10) \ln \frac{9}{11}}$$

$$T(10) = 35 - 55e^{5 \ln \frac{9}{11}}$$

$$T(10) = 35 - 55e^{\ln \left(\frac{9}{11}\right)^5}$$

$$T(10) = 35 - 55 \left(\frac{9}{11}\right)^5$$

$$T(10) \approx 15$$

The temperature of the ice cream after 10 minutes will be about $15°$.

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■ 4. The temperature in a room is 25°, while outside temperature is -20°. After 5 minutes with the door open, the room temperature dropped to 20°. What will the temperature be in the room after 15 minutes, and when will the room temperature drop to 5°?

Solution:

If we plug everything we know,

$T_0 = 25^\circ$ Initial temperature of the room

$T_a = -20^\circ$ Temperature of the environment

$T(5) = 20^\circ$ At $t = 5$ minutes, the room has cooled to 20°

If we plug everything we know into the Newton's Law of Cooling solution equation, we get

$$T(t) = T_a + (T_0 - T_a)e^{-kt}$$

$$T(t) = -20 + (25 - (-20))e^{-kt}$$

$$T(t) = -20 + (25 + 20)e^{-kt}$$

$$T(t) = -20 + 45e^{-kt}$$

Substitute the initial condition $T(5) = 20^\circ$,

$$20 = -20 + 45e^{-5k}$$

$$40 = 45e^{-5k}$$

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$$\frac{8}{9} = e^{-5k}$$

$$\ln \frac{8}{9} = -5k$$

$$k = -\frac{1}{5} \ln \frac{8}{9}$$

then plug k into the equation modeling temperature over time.

$$T(t) = -20 + 45e^{-\left(-\frac{1}{5} \ln \frac{8}{9}\right)t}$$

$$T(t) = -20 + 45e^{\frac{1}{5}t \ln \frac{8}{9}}$$

We want to find the temperature of the room after 15 minutes, so we'll evaluate at $t = 15$ to find $T(15)$.

$$T(15) = -20 + 45e^{\frac{1}{5}(15) \ln \frac{8}{9}}$$

$$T(15) = -20 + 45e^{3 \ln \frac{8}{9}}$$

$$T(15) = -20 + 45e^{\ln \left(\frac{8}{9}\right)^3}$$

$$T(15) = -20 + 45 \left(\frac{8}{9}\right)^3$$

$$T(10) \approx 11.6^\circ$$

The room temperature after 15 minutes will be $11.6^\circ C$. To find the time t at which the room temperature drops to 5°, we'll substitute $T(t) = 5^\circ$.

$$5 = -20 + 45e^{\frac{1}{5}t \ln \frac{8}{9}}$$

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$$25 = 45e^{\frac{1}{5}t \ln \frac{8}{9}}$$

$$\frac{25}{45} = e^{\frac{1}{5}t \ln \frac{8}{9}}$$

$$\frac{5}{9} = e^{\frac{1}{5}t \ln \frac{8}{9}}$$

$$\ln \frac{5}{9} = \frac{1}{5}t \ln \frac{8}{9}$$

$$5 \ln \frac{5}{9} = t \ln \frac{8}{9}$$

$$t = \frac{5 \ln \frac{5}{9}}{\ln \frac{8}{9}}$$

$$t \approx 25$$

The room temperature drops to 5° after 25 minutes.

■ 5. A piece of metal is placed in a $20^\circ C$ room, and after 5 minutes the temperature of the metal is $120^\circ C$. After another 15 minutes the metal's temperature is $75^\circ C$. What was the initial temperature of the metal?

Solution:

If we plug everything we know,

$T_a = 20^\circ$ Temperature of the environment

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$T(5) = 120°$ At $t = 5$ minutes, the metal has cooled to $120°$

$T(20) = 75°$ At $t = 20$ minutes, the metal has cooled to $75°$

into the Newton's Law of Cooling solution equation, we get

$$T(t) = T_a + (T_0 - T_a)e^{-kt}$$

$$T(t) = 20 + (T_0 - 20)e^{-kt}$$

Substitute the conditions $T(5) = 120°$ and $T(20) = 75°$.

$$120 = 20 + (T_0 - 20)e^{-5k}$$

$$75 = 20 + (T_0 - 20)e^{-20k}$$

Simplify the system to

$$100 = (T_0 - 20)e^{-5k}$$

$$55 = (T_0 - 20)e^{-20k}$$

Divide the first equation by the second equation.

$$\frac{100}{55} = \frac{e^{-5k}}{e^{-20k}}$$

$$\frac{20}{11} = e^{15k}$$

$$\ln \frac{20}{11} = 15k$$

$$k = \frac{1}{15} \ln \frac{20}{11}$$

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Substitute this value for k into the first equation of the system.

$$100 = (T_0 - 20)e^{-5\left(\frac{1}{15}\ln\frac{20}{11}\right)}$$

$$100 = (T_0 - 20)e^{-\frac{1}{3}\ln\frac{20}{11}}$$

$$100 = (T_0 - 20)e^{\ln\left(\frac{20}{11}\right)^{-\frac{1}{3}}}$$

$$100 = (T_0 - 20)\left(\frac{20}{11}\right)^{-\frac{1}{3}}$$

$$100\left(\frac{20}{11}\right)^{\frac{1}{3}} = (T_0 - 20)$$

$$T_0 = 100\left(\frac{20}{11}\right)^{\frac{1}{3}} + 20$$

$$T_0 \approx 142°C$$

The initial temperature of the metal was approximately $142°C$.

■ 6. Room temperature can only be measured at temperatures less than or equal to $25°C$. We can't currently measure room temperature, which means the room must be warmer than $25°C$. Meanwhile, outside temperature is $-5°C$. After 15 minutes with the window open, room temperature has dropped to $20°C$, and to $17°C$ after another 5 minutes. What was the initial temperature of the room?

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Solution:

If we plug everything we know,

$T_a = -5°$ Temperature of the environment

$T(15) = 20°$ At $t = 15$ minutes, the room has cooled to $20°$

$T(20) = 17°$ At $t = 20$ minutes, the room has cooled to $17°$

into the Newton's Law of Cooling solution equation, we get

$$T(t) = T_a + (T_0 - T_a)e^{-kt}$$

$$T(t) = -5 + (T_0 + 5)e^{-kt}$$

Substitute the conditions $T(15) = 20°$ and $T(20) = 17°$.

$$20 = -5 + (T_0 + 5)e^{-15k}$$

$$17 = -5 + (T_0 + 5)e^{-20k}$$

Simplify the system to

$$25 = (T_0 + 5)e^{-15k}$$

$$22 = (T_0 + 5)e^{-20k}$$

Divide the first equation by the second equation.

$$\frac{25}{22} = \frac{e^{-15k}}{e^{-20k}}$$

$$\frac{25}{22} = e^{5k}$$

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$$\ln \frac{25}{22} = 5k$$

$$k = \frac{1}{5} \ln \frac{25}{22}$$

Substitute this value for k into the first equation of the system.

$$25 = (T_0 + 5)e^{-15\left(\frac{1}{5} \ln \frac{25}{22}\right)}$$

$$25 = (T_0 + 5)e^{-3 \ln \frac{25}{22}}$$

$$25 = (T_0 + 5)e^{\ln\left(\frac{25}{22}\right)^{-3}}$$

$$25 = (T_0 + 5)\left(\frac{25}{22}\right)^{-3}$$

$$25\left(\frac{25}{22}\right)^{3} = T_0 + 5$$

$$T_0 = 25\left(\frac{25}{22}\right)^{3} - 5$$

$$T_0 \approx 31.7°C$$

The initial temperature of the room was approximately $31.7°C$.

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ELECTRICAL SERIES CIRCUITS

■ 1. If a battery supplies a constant voltage of 9 V, has an inductance of 3 h and a resistance of $3\ \Omega$, and assuming $i(0) = 0$, find the current after 120 seconds.

Solution:

This series circuit doesn't have a capacitor, which means we can remove that term from Kirchhoff's second law.

$$L\frac{di}{dt} + Ri + \frac{1}{C}q = E(t)$$

$$L\frac{di}{dt} + Ri = E(t)$$

Plugging in what we know, we get

$$3\frac{di}{dt} + 3i = 9$$

$$i' + i = 3$$

This is a linear differential equation with integrating factor

$$I(t) = e^{\int P\ dt}$$

$$I(t) = e^{\int 1\ dt}$$

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$$I(t) = e^t$$

Therefore, we can rewrite the linear equation as

$$e^t \frac{di}{dt} + e^t i(t) = 3e^t$$

$$\frac{d}{dt}(e^t i(t)) = 3e^t$$

$$\int \frac{d}{dt}(e^t i(t))\ dt = \int 3e^t\ dt$$

$$e^t i(t) = 3e^t + C$$

$$i(t) = 3 + Ce^{-t}$$

Substituting the initial condition $i(0) = 0$, we find

$$0 = 3 + Ce^0$$

$$C = -3$$

So we can write the equation modeling the current i over time t as

$$i(t) = 3 - 3e^{-t}$$

Assuming time is measured in minutes, the current after 120 seconds is

$$i(2) = 3 - 3e^{-120}$$

$$i(2) = 3$$

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■ 2. Given $L = 10$ h, $R = 70\ \Omega$, $C = 0.01$ f, $E(t) = 0$, $q(0) = 0$, and $i(0) = 3$, find the charge $q(t)$ on the capacitor.

Solution:

Plugging what we know into the second order equation given by Kirchhoff's second law, we get

$$L\frac{d^2q}{dt^2} + R\frac{dq}{dt} + \frac{1}{C}q = E(t)$$

$$10q'' + 70q' + \frac{1}{0.01}q = 0$$

$$q'' + 7q' + 10q = 0$$

The associated characteristic equation for this second order homogeneous equation is

$$r^2 + 7r + 10 = 0$$

$$(r + 5)(r + 2) = 0$$

$$r = -5,\ -2$$

Because we found distinct real roots, we know the circuit is overdamped. With distinct real roots, we can say that the general solution and its derivative are given by

$$q(t) = c_1e^{-5t} + c_2e^{-2t}$$

$$q'(t) = i(t) = -5c_1e^{-5t} - 2c_2e^{-2t}$$

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Substituting the initial conditions $q(0) = 3$ and $i(0) = 0$ into these equations, we get

$$0 = c_1 e^{-5(0)} + c_2 e^{-2(0)}$$

$$0 = c_1 + c_2$$

$$c_2 = -c_1$$

and

$$3 = -5c_1 e^{-5(0)} - 2c_2 e^{-2(0)}$$

$$3 = -5c_1 - 2c_2$$

Substituting $c_2 = -c_1$ into the second equation gives

$$3 = -5c_1 + 2c_1$$

$$3 = -3c_1$$

$$c_1 = -1$$

Plugging this back into the equation for c_2, we get

$$c_2 = -(-1)$$

$$c_2 = 1$$

So the general solution can be written as

$$q(t) = -e^{-5t} + e^{-2t}$$

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■ 3. An inductance of 2 h is connected in series with a resistance of $10\ \Omega$ and a battery giving $E(t) = 120$ V. Initially, the current is zero. Formulate and solve an initial value problem modeling the electrical circuit.

Solution:

This series circuit doesn't have a capacitor, which means we can remove that term from Kirchhoff's second law.

$$L\frac{di}{dt} + Ri + \frac{1}{C}q = E(t)$$

$$L\frac{di}{dt} + Ri = E(t)$$

Plugging in what we know, we get

$$2\frac{di}{dt} + 10i = 120$$

$$i' + 5i = 60$$

This is a linear differential equation with integrating factor

$$I(t) = e^{\int P\ dt}$$

$$I(t) = e^{\int 5\ dt}$$

$$I(t) = e^{5t}$$

Therefore, we can rewrite the linear equation as

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$$e^{5t}\frac{di}{dt} + 5e^{5t}i(t) = 60e^{5t}$$

$$\frac{d}{dt}(e^{5t}i(t)) = 60e^{5t}$$

$$\int \frac{d}{dt}(e^{5t}i(t))\ dt = \int 60e^{5t}\ dt$$

$$e^{5t}i(t) = 12e^{5t} + C$$

$$i(t) = 12 + Ce^{-5t}$$

Substituting the initial condition $i(0) = 0$, we find

$$0 = 12 + Ce^{0}$$

$$C = -12$$

So we can write the equation modeling the current i over time t as

$$i(t) = 12 - 12e^{-5t}$$

■ 4. Suppose $L = 2$ h, $R = 6\ \Omega$, $E = 24e^{3t}$ V, and $i(0) = 0$. Formulate and solve an initial value problem that models the electrical circuit.

Solution:

This series circuit doesn't have a capacitor, which means we can remove that term from Kirchhoff's second law.

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$$L\frac{di}{dt} + Ri + \frac{1}{C}q = E(t)$$

$$L\frac{di}{dt} + Ri = E(t)$$

Plugging in what we know, we get

$$2\frac{di}{dt} + 6i = 24e^{3t}$$

$$i' + 3i = 12e^{3t}$$

This is a linear differential equation with integrating factor

$$I(t) = e^{\int P\ dt}$$

$$I(t) = e^{\int 3\ dt}$$

$$I(t) = e^{3t}$$

Therefore, we can rewrite the linear equation as

$$e^{3t}\frac{di}{dt} + 3e^{3t}i(t) = 12e^{3t}e^{3t}$$

$$\frac{d}{dt}(e^{3t}i(t)) = 12e^{6t}$$

$$\int \frac{d}{dt}(e^{3t}i(t))\ dt = \int 12e^{6t}\ dt$$

$$e^{3t}i(t) = 2e^{6t} + C$$

$$i(t) = 2e^{3t} + Ce^{-3t}$$

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Substituting the initial condition $i(0) = 0$, we find

$$0 = 2e^0 + Ce^0$$

$$C = -2$$

So we can write the equation modeling the current i over time t as

$$i(t) = 2e^{3t} - 2e^{-3t}$$

■ 5. Suppose $L = 2$ h, $R = 40\ \Omega$, $C = 0.005$ f, $E = 100$ V, $q(0) = 9$ C, and $q'(0) = i(0) = 0$. Formulate and solve an initial value problem that models this LRC circuit.

Solution:

Plugging what we know into the second order equation given by Kirchhoff's second law, we get

$$L\frac{d^2q}{dt^2} + R\frac{dq}{dt} + \frac{1}{C}q = E(t)$$

$$2q'' + 40q' + \frac{1}{0.005}q = 100$$

$$q'' + 20q' + 100q = 50$$

The associated characteristic equation for this second order homogeneous equation is

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$$r^2 + 20r + 100 = 0$$

$$(r + 10)(r + 10) = 0$$

$$r = -10, \ -10$$

Because we found equal real roots, we know the circuit is critically damped. With equal real roots, we can say that the complementary solution and its derivative are given by

$$q_c(t) = c_1 e^{-10t} + c_2 t e^{-10t}$$

$$q_c'(t) = i(t) = -10c_1 e^{-10t} + c_2 e^{-10t} - 10tc_2 e^{-10t}$$

Substituting the initial conditions $q(0) = 9$ and $i(0) = 0$ into these equations, we get

$$9 = c_1 e^{-10(0)} + c_2(0) e^{-10(0)}$$

$$c_1 = 9$$

and

$$0 = -10c_1 e^{-10(0)} + c_2 e^{-10(0)} - 10(0)c_2 e^{-10(0)}$$

$$0 = -10c_1 + c_2$$

$$c_2 = -10c_1$$

$$c_2 = -10(9)$$

$$c_2 = -90$$

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So the complementary solution can be written as

$$q_c(t) = 9e^{-10t} - 90te^{-10t}$$

Our guess for the particular solution will be

$$y_p(x) = A$$

Taking the first and second derivatives of this guess, we get

$$y_p'(x) = 0$$

$$y_p''(x) = 0$$

Plugging the first two derivatives into the original differential equation, we get

$$0 + 20(0) + 100A = 50$$

$$A = \frac{1}{2}$$

$$y_p(x) = \frac{1}{2}$$

Putting this particular solution together with the complementary solution gives us the general solution to the differential equation.

$$q(t) = 9e^{-10t} - 90te^{-10t} + \frac{1}{2}$$

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■ 6. An LRC circuit is set up with an inductance of $1/2$ h, a resistance of $1\ \Omega$, and a capacitance of $2/5$ f. Assuming the initial charge is 2 C and the initial current is 6 A, find the solution function describing the charge on the capacitor at any time t.

Solution:

Plugging what we know into the second order equation given by Kirchhoff's second law, we get

$$L\frac{d^2q}{dt^2} + R\frac{dq}{dt} + \frac{1}{C}q = E(t)$$

$$\frac{1}{2}q'' + q' + \frac{5}{2}q = 0$$

$$q'' + 2q' + 5q = 0$$

The associated characteristic equation for this second order homogeneous equation is

$$r^2 + 2r + 5 = 0$$

$$r = -1 \pm 2i$$

Because we found complex conjugate roots, we know the circuit is underdamped. With complex roots, we can say that the general solution and its derivative are given by

$$q(t) = e^{-t}(c_1\cos(2t) + c_2\sin(2t))$$

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$$q'(t) = i(t) = -e^{-t}(c_1\cos(2t) + c_2\sin(2t)) + e^{-t}(-2c_1\sin(2t) + 2c_2\cos(2t))$$

$$= e^{-t}((2c_2 - c_1)\cos(2t) - (c_2 + 2c_1)\sin(2t))$$

Substituting the initial conditions $q(0) = 2$ and $i(0) = 6$ into these equations, we get

$$2 = e^{-(0)}(c_1\cos(2(0)) + c_2\sin(2(0)))$$

$$2 = c_1$$

and

$$6 = e^{-0}((2c_2 - c_1)\cos(2(0)) - (c_2 + 2c_1)\sin(2(0)))$$

$$6 = 2c_2 - c_1$$

$$6 = 2c_2 - 2$$

$$c_2 = 4$$

So the general solution can be written as

$$q(t) = e^{-t}(2\cos(2t) + 4\sin(2t))$$

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SPRING AND MASS SYSTEMS

■ 1. Find the movement of a 6 kg mass attached to a spring with spring constant $k = 3$ kg/s^2 and damping constant $\beta = 6$ kg/s, given $x(0) = 2$ and $x'(0) = 0$.

Solution:

The differential equation for this system is

$$\frac{d^2x}{dt^2} + 2\lambda\frac{dx}{dt} + \omega^2 x = 0$$

with $m = 6$, $k = 3$, and $\beta = 6$, where $2\lambda = \beta/m$ and $\omega^2 = k/m$. The value of λ is given by

$$2\lambda = \frac{\beta}{m}$$

$$\lambda = \frac{\beta}{2m}$$

$$\lambda = \frac{6}{2(6)} = \frac{1}{2}$$

and ω^2 is given by

$$\omega^2 = \frac{k}{m}$$

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$$\omega^2 = \frac{3}{6} = \frac{1}{2}$$

The roots of the associated characteristic equation, $r^2 + 2\lambda r + \omega^2 = 0$, are

$$r_{1,2} = -\lambda \pm \sqrt{\lambda^2 - \omega^2}$$

$$r_{1,2} = -\frac{1}{2} \pm \sqrt{\frac{1}{4} - \frac{1}{2}}$$

$$r_{1,2} = -\frac{1}{2} \pm i\frac{1}{2}$$

Because we find roots in which $\lambda^2 - \omega^2 < 0$, the spring and mass system is underdamped and the general solution and its derivative are

$$x(t) = e^{-\lambda t}(c_1 \cos\sqrt{\omega^2 - \lambda^2}t + c_2 \sin\sqrt{\omega^2 - \lambda^2}t)$$

$$x(t) = e^{-\frac{1}{2}t}\left(c_1 \cos\frac{1}{2}t + c_2 \sin\frac{1}{2}t\right)$$

$$x'(t) = -\frac{1}{2}e^{-\frac{1}{2}t}\left(c_1 \cos\frac{t}{2} + c_2 \sin\frac{t}{2}\right) + e^{-\frac{1}{2}t}\left(-\frac{1}{2}c_1 \sin\frac{t}{2} + \frac{1}{2}c_2 \cos\frac{t}{2}\right)$$

$$= e^{-\frac{1}{2}t}\left(\left(-\frac{1}{2}c_1 + \frac{1}{2}c_2\right)\cos\frac{t}{2} + \left(-\frac{1}{2}c_1 - \frac{1}{2}c_2\right)\sin\frac{t}{2}\right)$$

The question states that the initial position is $x(0) = 2$ and the initial velocity is $x'(0) = 0$, so we'll plug these into $x(t)$ and $x'(t)$, and we get

$$2 = e^{-\frac{1}{2}(0)}\left(c_1 \cos\frac{1}{2}(0) + c_2 \sin\frac{1}{2}(0)\right)$$

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$$c_1 = 2$$

and

$$0 = e^{-\frac{1}{2}(0)}\left(\left(-\frac{1}{2}(2) + \frac{1}{2}c_2\right)\cos\frac{0}{2} + \left(-\frac{1}{2}(2) - \frac{1}{2}c_2\right)\sin\frac{0}{2}\right)$$

$$0 = -1 + \frac{1}{2}c_2$$

$$c_2 = 2$$

So the equation modeling the motion of this spring and mass system, with these particular initial conditions for velocity and position, is given by

$$x(t) = e^{-\frac{t}{2}}\left(2\cos\frac{t}{2} + 2\sin\frac{t}{2}\right)$$

■ 2. Assume an object weighing 8 lbs stretches a spring 12 in. Find the equation of motion if the spring is released from equilibrium with an upward velocity of $\sqrt{2}$ ft/sec.

Solution:

First we'll use Hooke's Law to find the spring constant k.

$$F = ks$$

$$8 = k\,(1)$$

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$$k = 8$$

Now we'll use $W = mg$ to convert the weight into mass.

$$W = mg$$

$$8 = m(32)$$

$$m = \frac{1}{4}$$

Plugging everything we have into the second order equation, we get

$$\frac{d^2x}{dt^2} + \frac{k}{m}x = 0$$

$$\frac{d^2x}{dt^2} + \frac{8}{\frac{1}{4}}x = 0$$

$$\frac{d^2x}{dt^2} + 32x = 0$$

From this equation, we see that $\omega = \sqrt{32} = 4\sqrt{2}$, which means that the general solution and its derivative are

$$x(t) = c_1 \cos(\omega t) + c_2 \sin(\omega t)$$

$$x(t) = c_1 \cos(4\sqrt{2}t) + c_2 \sin(4\sqrt{2}t)$$

and

$$x'(t) = -4\sqrt{2}c_1 \sin(4\sqrt{2}t) + 4\sqrt{2}c_2 \cos(4\sqrt{2}t)$$

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The question states that the initial position is $x(0) = 0$ and the initial velocity is $x'(0) = -\sqrt{2}$, so we'll plug these into $x(t)$ and $x'(t)$, and we get

$$0 = c_1 \cos(4\sqrt{2}(0)) + c_2 \sin(4\sqrt{2}(0))$$

$$c_1 = 0$$

and

$$-\sqrt{2} = -4\sqrt{2}(0)\sin(4\sqrt{2}(0)) + 4\sqrt{2}c_2 \cos(4\sqrt{2}(0))$$

$$c_2 = -\frac{1}{4}$$

So the equation modeling the motion of this spring and mass system, with these particular initial conditions for velocity and position, is given by

$$x(t) = (0)\cos(4\sqrt{2}t) - \frac{1}{4}\sin(4\sqrt{2}t)$$

$$x(t) = -\frac{1}{4}\sin(4\sqrt{2}t)$$

■ 3. An 8 lb weight is attached to an 11 ft spring. When the mass comes to rest in the equilibrium position, the spring measures 15 ft. The system is immersed in a medium that imparts a damping force equal to $3/2$ times the instantaneous velocity of the mass. Find the equation of motion if the mass is pushed upward from the equilibrium position with an initial upward velocity of 4 ft/sec. What is the position of the mass after 15 sec?

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Solution:

Hooke's Law tells us that the spring constant is

$$F = ks$$

$$8 = k(15 - 11)$$

$$8 = k(4)$$

$$k = 2 \text{ lb/ft}$$

Now we'll use $W = mg$ to convert the weight into mass.

$$W = mg$$

$$8 = m(32)$$

$$m = \frac{1}{4} \text{ slug}$$

With the spring constant and the mass, and $\beta = 3/2$ since the damping force is $3/2$ times the instantaneous velocity, we can plug everything into the differential equation.

$$\frac{d^2x}{dt^2} + \frac{\beta}{m}\left(\frac{dx}{dt}\right) + \frac{k}{m}x = 0$$

$$\frac{d^2x}{dt^2} + \frac{\frac{3}{2}}{\frac{1}{4}}\left(\frac{dx}{dt}\right) + \frac{2}{\frac{1}{4}}x = 0$$

$$\frac{d^2x}{dt^2} + 6\left(\frac{dx}{dt}\right) + 8x = 0$$

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The associated characteristic equation and its roots are

$$r^2 + 6r + 8 = 0$$

$$r = -4, -2$$

Because we find roots in which $\lambda^2 - \omega^2 > 0$, the spring and mass system is overdamped and the general solution and its derivative are

$$x(t) = c_1 e^{-4t} + c_2 e^{-2t}$$

and

$$x'(t) = -4c_1 e^{-4t} - 2c_2 e^{-2t}$$

The question states that the initial position is $x(0) = 0$ and the initial velocity is $x'(0) = -4$, so we'll plug these into $x(t)$ and $x'(t)$, and we get

$$0 = c_1 e^{-4(0)} + c_2 e^{-2(0)}$$

$$c_1 = -c_2$$

and

$$-4 = -4c_1 e^{-4(0)} - 2c_2 e^{-2(0)}$$

$$-4 = -4c_1 - 2c_2$$

$$-4 = 4c_2 - 2c_2$$

$$-4 = 2c_2$$

$$c_2 = -2$$

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$$c_1 = 2$$

So the equation modeling the motion of this spring and mass system, with these particular initial conditions for velocity and position, is given by

$$x(t) = 2e^{-4t} - 2e^{-2t}$$

After 15 seconds the mass is at a position of

$$x(15) = 2e^{-4(15)} - 2e^{-2(15)} \approx -1.9 \times 10^{-13} \approx 0$$

so it's at equilibrium.

■ 4. A 2 kg mass stretches a spring with a length of 20 cm. The system is attached to a dashpot that creates a damping force equal to 28 times the instantaneous velocity of the mass. Find the motion equation if the mass is released from rest at a point 6 cm below equilibrium.

Solution:

Hooke's Law tells us that the spring constant is

$$F = ks$$

$$2(9.8) = k(0.2)$$

$$k = 98 \text{ lb/ft}$$

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With the spring constant and the mass, and $\beta = 28$ since the damping force is 28 times the instantaneous velocity, we can plug everything into the differential equation.

$$\frac{d^2x}{dt^2} + \frac{\beta}{m}\left(\frac{dx}{dt}\right) + \frac{k}{m}x = 0$$

$$\frac{d^2x}{dt^2} + \frac{28}{2}\left(\frac{dx}{dt}\right) + \frac{98}{2}x = 0$$

$$\frac{d^2x}{dt^2} + 14\left(\frac{dx}{dt}\right) + 49x = 0$$

The associated characteristic equation and its roots are

$$r^2 + 14r + 49 = 0$$

$$r = -7, -7$$

Because we find roots in which $\lambda^2 - \omega^2 = 0$, the spring and mass system is critically damped and the general solution and its derivative are

$$x(t) = e^{-\lambda t}(c_1 + c_2 t)$$

$$x(t) = e^{-7t}(c_1 + c_2 t)$$

and

$$x'(t) = -7e^{-7t}c_1 - 7te^{-7t}c_2 + e^{-7t}c_2$$

The question states that the initial position is $x(0) = 6$ and the initial velocity is $x'(0) = 0$, so we'll plug these into $x(t)$ and $x'(t)$, and we get

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$$6 = e^{-7(0)}(c_1 + c_2(0))$$

$$c_1 = 6$$

and

$$0 = -7e^{-7(0)}c_1 - 7(0)e^{-7(0)}c_2 + e^{-7(0)}c_2$$

$$0 = -7c_1 + c_2$$

$$0 = -7(6) + c_2$$

$$c_2 = 42$$

So the equation modeling the motion of this spring and mass system, with these particular initial conditions for velocity and position, is given by

$$x(t) = e^{-7t}(6 + 42t)$$

■ 5. A 16 lb weight stretches a spring 2 ft. Assume the damping force on the system is equal to the instantaneous velocity of the mass. Find the equation of motion of the mass.

Solution:

Hooke's Law tells us that the spring constant is

$$F = ks$$

$$16 = k(2)$$

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$$k = 8 \text{ lb/ft}$$

Now we'll use $W = mg$ to convert the weight into mass.

$$W = mg$$

$$16 = m(32)$$

$$m = \frac{1}{2}$$

With the spring constant and the mass, and $\beta = 1$ since the damping force is equal to the instantaneous velocity, we can plug everything into the differential equation.

$$\frac{d^2x}{dt^2} + \frac{\beta}{m}\left(\frac{dx}{dt}\right) + \frac{k}{m}x = 0$$

$$\frac{d^2x}{dt^2} + \frac{1}{\frac{1}{2}}\left(\frac{dx}{dt}\right) + \frac{8}{\frac{1}{2}}x = 0$$

$$\frac{d^2x}{dt^2} + 2\left(\frac{dx}{dt}\right) + 16x = 0$$

The associated characteristic equation and its roots are

$$r^2 + 2r + 16 = 0$$

$$r = -1 \pm i\sqrt{15}$$

Because we find roots in which $\lambda^2 - \omega^2 < 0$, the spring and mass system is underdamped and the general solution is

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$$x(t) = e^{-\lambda t}(c_1 \cos\sqrt{\omega^2 - \lambda^2}t + c_2 \sin\sqrt{\omega^2 - \lambda^2}t)$$

$$x(t) = e^{-t}(c_1 \cos\sqrt{15}t + c_2 \sin\sqrt{15}t)$$

■ 6. A mass of 9.8 kg stretches a spring 2.45 m. Find the equation that models the motion of the mass if we release the mass when $t = 0$ from a position 5 m above equilibrium, with a downward velocity of 2 m/s.

Solution:

First we'll use Hooke's Law to find the spring constant k.

$$F = ks$$

$$9.8 = k(2.45)$$

$$k = 4$$

Now we'll use $W = mg$ to convert the weight into mass.

$$W = mg$$

$$9.8 = m(9.8)$$

$$m = 1$$

Plugging everything we have into the second order equation, we get

$$\frac{d^2x}{dt^2} + \frac{k}{m}x = 0$$

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$$\frac{d^2x}{dt^2} + \frac{4}{1}x = 0$$

$$\frac{d^2x}{dt^2} + 4x = 0$$

From this equation, we see that $\omega = \sqrt{4} = 2$, which means that the general solution and its derivative are

$$x(t) = c_1 \cos(\omega t) + c_2 \sin(\omega t)$$

$$x(t) = c_1 \cos(2t) + c_2 \sin(2t)$$

and

$$x'(t) = -2c_1 \sin(2t) + 2c_2 \cos(2t)$$

The question states that the initial position is $x(0) = -5$ and the initial velocity is $x'(0) = 2$, so we'll plug these into $x(t)$ and $x'(t)$, and we get

$$-5 = c_1 \cos(2(0)) + c_2 \sin(-2(0))$$

$$c_1 = -5$$

and

$$2 = -2c_1 \sin(2(0)) + 2c_2 \cos(2(0))$$

$$c_2 = 1$$

So the equation modeling the motion of this spring and mass system, with these particular initial conditions for velocity and position, is given by

$$x(t) = -5 \cos(2t) + \sin(2t)$$

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